SCOUT: Supplemental Material

S1 Evaluation Setup: Supplementary Details

This section consolidates setup details referenced from §5.1, including synthetic workload construction, operating-point selection, and scale and microbenchmark details.

S1.1 Synthetic Workload Construction

The synthetic evaluation uses four workload families spanning concentrated, diffuse, and heavy-tail noise workloads. All workloads are generated over the same shared segment space; drift is controlled independently by the side-imbalance parameter q.

Bimodal and diffuse `bimodal` concentrates all differences in B dirty segments, with the remaining segments clean. `diffuse` spreads differences approximately uniformly across a large number of segments.

Gray-zone heavy-tail workloads `heavy_share` and `mix` place a large difference count d_{big} on a small set of segments while many background segments each receive a smaller count d_{small}. For `mix`, difficulty is controlled by two axes: noise-to-signal ratio `S/B` and separability `ratio`$=d_{\text{big}}/d_{\text{small}}$. `heavy_share` is the high-multiplier special case used to probe behavior near the ranking-collapse boundary.

Recall target For mixed workloads, `recall_big` denotes the fraction of segments carrying d_{big} that are covered by the candidate set. This metric isolates whether the method recovers the operationally important heavy segments when background noise is present.

S1.2 Metrics and Operating-Point Protocol

At each stress point, every method sweeps its own internal control knob: SCOUT sweeps truncation ratio `top_r`, B2 sweeps top-K, and B3 sweeps drill-down aggressiveness. The operating point is the minimum candidate fraction satisfying the recall requirement `median(recall_big)` ≥ 0.95. If no setting satisfies the recall floor within the 2% candidate-budget cap, the point is recorded as **N/A**.

S1.3 Scale and Microbenchmark Details

Synthetic experiments sweep total segment count from 10^4 to 10^7. The YCSB validation replays up to $\sim 2\times 10^9$ records across P=100,000 range-partitioned segments, producing nearly 50× hot-to-cold segment-size skew under the Zipfian workload.

Cost item	Reference anchor point
Exchange (one-sided digest)	$4VLm_0$ bytes; V=2: 512 KiB, V=3: 768 KiB
Decode (stats + score)	∼0.33 ms/view; ∼1 ms at V=3; independent of P
Build (per-key update)	∼2.9 μs/key; includes window lookups and k hash updates

Table S1: Reference anchor points for probe overhead (L=2048, m_0=32, k=2). Decode and Build are measured from a C++ microbenchmark.

At this scale, the probe's digest-side overhead remains small: one-sided digest size stays below 1 MiB and decode time stays below 1 ms; reference anchor points for exchange size, decode latency, and build cost are summarized in Table S1. In a single-threaded microbenchmark on a Xeon Platinum 8468, each key update takes approximately 2.9 μs; building the probe over 10^9 keys therefore completes in minutes, with the dominant wall-clock cost still determined by local sequential-scan I/O rather than probe computation.

S2 Spatial Coupling: Supplementary Results

This section supplements the implementation details and additional results for §5.2, including triage strategy definitions and their combinations, the complete numerical results and

interpretation of the window-width W ablation, and halo suppression analysis.

S2.1 Derivation Details for the Candidate-Set Size Expectation

This section provides the detailed derivation behind Equations (1) and (2) in the main text. Consider the section axis of a single view as a linear interval (without circular wrap-around), with window width W and window start positions approximately uniform over $[1, L-W+1]$. If signals are separable, the top-t truncation covers the high-energy windows of all dirty segments. A clean segment is falsely included in the candidate set in a single view if and only if its window overlaps on the section axis with any selected window. For two windows of width W, overlap occurs if and only if their start positions differ by less than W; each selected window therefore "contaminates" a neighborhood of width $2W-1$ on the section axis.

When $t(2W-1) \ll L$ so that these contaminated neighborhoods are approximately non-overlapping, the probability that a clean segment's window start falls within a contaminated region is

$$p_{\text{halo}} \approx \frac{t(2W-1)}{L},$$

which is Equation (2) in the main text. After intersecting candidate sets across V approximately independent views, the probability that a clean segment is falsely included in every view is approximately p_{halo}^{V}; multiplying by the total number of clean segments $P-B$ gives the expected false-inclusion volume, while the contribution from truly dirty segments is B, yielding Equation (1).

S2.2 Break-Even Condition for W

The preceding section established the relationship between p_{halo} and (t, W, L). In practice, t is not a free parameter but is determined by the minimum truncation count $t_{\text{op}}(W)$ required for target recall. Substituting into Equation (2) gives the operating false-inclusion probability

$$p_{\text{halo,op}}(W) \triangleq p_{\text{halo}}\big(W \mid t_{\text{op}}(W)\big) \approx \frac{t_{\text{op}}(W)\,(2W-1)}{L}.$$

Whether $W > 1$ yields a net benefit therefore depends on whether the compression of t_{op} through window aggregation outweighs the geometric penalty. Taking $W=1$ as the baseline, the necessary condition for a net benefit (i.e., $p_{\text{halo,op}}(W) < p_{\text{halo,op}}(1)$) is

$$t_{\text{op}}(W) < \frac{t_{\text{op}}(1)}{2W-1}. \tag{S1}$$

Equation (S1) captures the opposition between two forces: wider windows compress $t_{\text{op}}(W)$ through matched filtering, but each selected window contaminates a larger halo neighborhood by the factor $(2W-1)$. A net benefit requires the compression to outpace this geometric penalty.

Under high collision pressure (P/L large), contiguous-window aggregation (§5.2) significantly stabilizes the energy ranking, reducing the redundancy of t_{op} relative to the ideal lower bound BW. In this regime, the compression is sufficient to offset the halo penalty, and $W=2$ achieves a net benefit. Conversely, when L/B is small and dirty windows are already dense on the section axis, $t_{\text{op}}(1)$ is close to the minimum needed to cover all B segments. Little compression headroom remains for larger W, the geometric penalty dominates, and candidate cost rebounds—as observed for $W=4, 8$ at $B=100$, $L=2048$ in Table S5.

S2.3 Triage Strategy Definitions and Combinations

We compare four triage strategies, summarized in Table S2. The core idea of `hotcold` filtering is to use relatively low-energy regions as a weak negative signal, helping suppress collision-induced false positives among surviving candidates after hot-region localization.

Specifically, for the section energy sequence $\{E_s\}$ in each view, the t sections with the highest energy form the hot set H, and the t sections with the lowest energy form the cold set C. For any window start position u, we count how many covered sections fall in H and C, respectively:

$$h(u) = |\{u, \ldots, u+W-1\} \cap H|,\ c(u) = |\{u, \ldots, u+W-1\} \cap C|.$$

For segment p with window start $u = h_v(p)$, $h(u)$ and $c(u)$ are its hot- and cold-zone counts.

Strategy	Description
`section_topr`	Select top-r sections by energy; baseline without contiguity.
`window_topr`	Slide a window, sum energy, and select top-r windows; matched filtering via spatial coupling.
`hotcold_r`	Use both hot (high-energy) and cold (low-energy) counts to separate collision false positives from true differences.
`window_topr_hotcold`	Apply `window_topr` first, then `hotcold` filtering; combines spatial aggregation with halo suppression.

Table S2: Comparison of triage strategy combinations.

The filtering condition is $h(u) > c(u)$. The rationale is as follows: a genuinely dirty segment contributes significant energy across the W sections it covers, so those positions tend to enter the hot set, making $h(u)$ dominant; a false positive caused by hash collisions may overlap some hot sections, but

its remaining covered positions are more likely to stay in the lower tail of the energy ranking, increasing $c(u)$. Cold-set hits therefore act as a weak negative signal that helps suppress collision-induced false positives around genuine hot zones. Strategy `hotcold_r` directly retains windows satisfying this condition and reverse-maps them to candidates; `window_topr_hotcold` first selects top-r windows by energy and then applies the same filter.

S2.4 Complete Results for the Window-Width W Ablation

This section provides the full configuration and results for the W ablation experiment described in §5.2.

We sweep $W \in \{1,2,4,8,L\}$ under the configurations in Table S3, where $W = L$ serves as a negative control in which spatial locality is entirely destroyed.

Parameter	Values
W	$\{1,2,4,8,L\}$
(P,L)	$\{(10^4,2048),\ldots,(10^7,4096)\}$
B	$\{10,100\}$
Fixed	$V=2,\ k=3,\ m_0=32$, strategy `window_topr`

Table S3: Ablation configuration for window width W (fixed V=2, k=3, m_0=32, strategy `window_topr`).

Figure S1 summarizes the sweep of the minimum `cand_frac` across pressure configurations for $W \in \{1,2,4,8\}$. To make the trend concrete, Table S4 reports representative B=10 configurations and the corresponding savings of W=2 over W=1. Across these settings, a modest contiguous window (W=2) consistently matches or improves upon per-section ranking (W=1), and the gain becomes more pronounced as collision pressure P/L increases. At the highest-pressure point (P=10^7, L=2048), W=2 reduces candidate cost by approximately 3.8× relative to W=1. The benefit does not grow monotonically with W, however: when dirty windows already occupy a dense portion of the section axis, the compression of the operating top-r quota is no longer sufficient to offset the larger halo induced by wider windows, and reversal appears in dense low-L/B settings. Table S5 reports the full numerical values, and Supplement S2.2 analyzes the underlying trade-off.

Figure S3 isolates the negative-control boundary case W=L. Once the window spans the entire section axis, all segments share the same global window and spatial locality disappears. The reverse-mapped candidate set therefore degenerates toward the full segment universe, yielding `cand_cnt` $\approx P$ in the boundary regime. This behavior confirms that candidate compression depends on preserving locality on the section axis, rather than merely increasing the number of covered sections.

Pressure	$W=1$ `cand_frac`	$W=2$ `cand_frac`	$W=2$ vs. $W=1$ saving
$P=10^4, L=2048$	0.0013	0.0010	1.3×
$P=10^5, L=2048$	0.000245	0.000215	1.1×
$P=10^7, L=2048$	0.000397	0.000105	**3.8**×
$P=10^7, L=4096$	0.000176	0.000104	**1.8**×

Table S4: Minimum `cand_frac` under symmetric injection ($B=10$, fixed V=2, k=3, m_0=32).

Table S5 reports the full numerical values. For $B=10$, the benefit of $W>1$ is more pronounced at high collision pressure, whereas for $B=100$ the differences among W values narrow considerably.

P	L	$W=1$	$W=2$	$W=4$	$W=8$
$B=10$					
10^4	2048	0.0013	0.0011	0.0157	0.0111
10^4	4096	0.0031	0.0013	0.0010	0.0011
10^5	2048	0.00025	0.00019	0.01045	0.0101
10^5	4096	0.00026	0.00022	0.00018	0.0002
10^7	2048	0.0003674	0.0001054	0.0001137	0.0100871
10^7	4096	0.0001832	0.0001029	0.0001012	0.0001005
$B=100$					
10^4	2048	0.0266	0.0185	0.0474	0.0470
10^4	4096	0.0301	0.0259	0.0187	0.0195
10^5	2048	0.01152	0.01079	0.04097	0.04066
10^5	4096	0.01271	0.01141	0.01081	0.01073
10^7	2048	0.0125068	0.0100346	0.0400252	0.0401635
10^7	4096	0.0108632	0.0100551	0.010038	0.0100062

Table S5: Minimum `cand_frac` for $W \in \{1,2,4,8\}$ under two density levels ($B \in \{10,100\}$).

For a more intuitive comparison of the relative performance of each W value across configurations, Figure S2 presents a pairwise dominance-rate heatmap: each cell indicates the fraction of all pressure points at which the row method achieves a lower `cand_frac` than the column method.

S2.5 Energy Signal Coverage Verification

Figure 3 in the main text demonstrates the advantage of contiguous structure (`WindowStart`) over random grouping (`RandomSections`) through diff-capture curves. This section complements that comparison with energy signal coverage curves, verifying from the perspective of observable quantities whether "high-energy regions are more concentrated"—the intermediate mechanism. Figure S4 shows that under `WindowStart`, selecting the top-r fraction of high-energy regions captures a larger share of total energy; this corroborates the diff-capture curves in the main text.

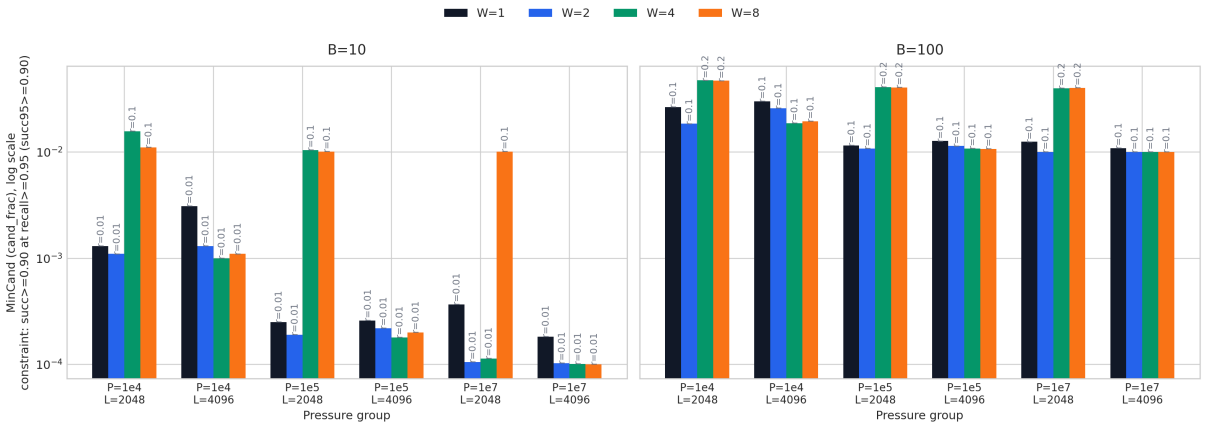


Figure S1: Window-width W ablation: minimum `cand_frac` required to meet the reliability threshold across pressure configurations (log scale; lower is better).

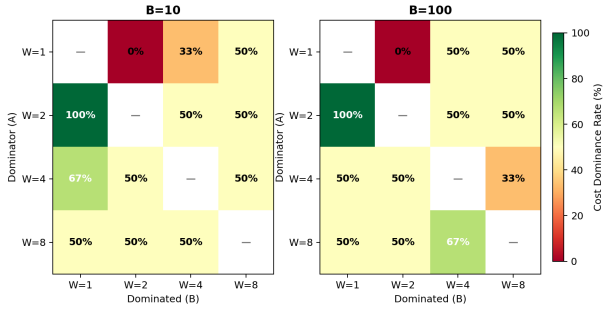


Figure S2: Pairwise dominance-rate heatmap across pressure configurations. Cell (i, j) shows the fraction of all pressure points at which $W = i$ achieves a lower minimum `cand_frac` than $W = j$ (ties count as 0.5). Greener cells indicate that the row W wins more frequently.

S2.6 Halo Suppression via Cold-Zone Signals

Following the definition in the main text, `halo` = `cand_frac` $\cdot (P/B)$ represents the candidate multiplier—the factor by which the candidate set exceeds the B dirty segments that must be covered. Under the reliability threshold (`succ95` ≥ 0.90, meaning at least 95% recall in at least 90% of trials), we select the minimum candidate budget that achieves the target recall and take the more conservative result across the two sides.

Strategy comparison Table S6 presents the best operating-point comparison across strategies under the same budget framework at $B = 100$. The combined strategy `window_topr_hotcold` reduces the halo to 4.0×, a roughly 5.5× improvement over the baseline that does not exploit contiguity (`section_topr`, 22.0×). Additionally, `hotcold_r` alone (16.4×) performs far worse than the combination with window-based localization, indicating that cold-zone signals are better suited as a second-stage filter after contiguous-window localization, rather than as a standalone first-stage localization mechanism.

Strategy	Halo factor	Relative to baseline
`section_topr`	**22.0**×	(baseline)
`window_topr`	4.2×	5.2× better
`hotcold_r`	16.4×	1.3× better
`window_topr_hotcold`	**4.0**×	**5.5× better**

Table S6: $B = 100$: best operating-point comparison (same budget; under reliability constraint).

B	Best strategy	`halo` (interpretation)
10	`window_topr_hotcold`	**1.0** (candidate size on the order of B)
100	`window_topr_hotcold`	**4.0** (still better than baseline's 22)

Table S7: Representative halo values as dirty-segment density varies.

Halo variation with dirty-segment density Table S7 shows the halo level of the best strategy under two density levels. When dirty segments are sparse ($B = 10$), `window_topr_hotcold` achieves a halo of approximately 1.0, meaning the candidate-set size is of the same order as the number of truly dirty segments and downstream verification incurs virtually no extra overhead. When dirty segments are denser ($B = 100$), the halo rises to approximately 4.0, consistent with the increased overlap of energy peaks on the section axis and higher collision pressure. Nonetheless, the combined strategy remains significantly better than all alternatives at both density levels. This reduction in halo cost under the combined strategy is consistent with the controlled candidate

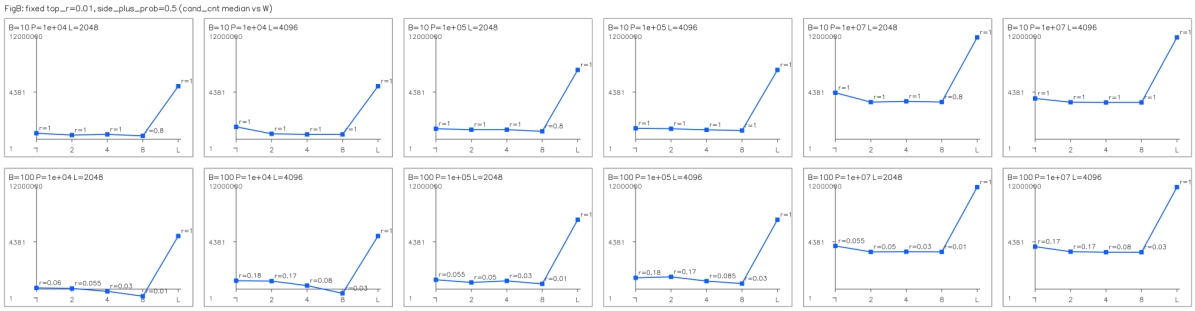


Figure S3: Absolute candidate count `cand_cnt` as a function of W under a fixed truncation budget (`top_r`=0.01). The rightmost point in each curve corresponds to $W{=}L$ (negative control).

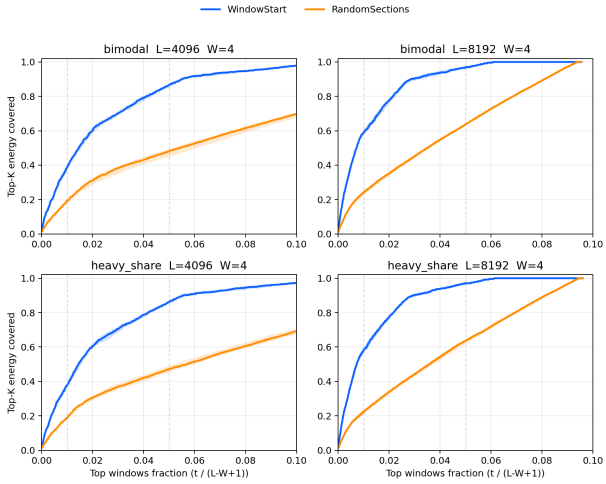


Figure S4: Top-K energy coverage. x-axis: top-r; y-axis: cumulative energy share. The `WindowStart` curve is steeper.

behavior observed in §5.4.

S3 Drift Robustness: Supplementary Details

This section provides cross-configuration validation of the drift robustness conclusions established in §5.3. The main text demonstrated that de-drifted energy remains stable across both the sizing and gating dimensions under varying drift conditions, based on a single set of probe parameters. Here, we extend that validation to a broader range of probe parameters and fault pressures, and additionally verify that triage accuracy is unaffected by the choice of statistic, justifying the use of de-drifted energy throughout the pipeline.

Table S8 summarizes the full experimental setup. The base configuration corresponds to the primary experiment in §5.3; two extended configurations are used for cross-configuration validation. The three score types correspond to the statistics evaluated in the experiments: `dedrift_countscore` is the de-drifted energy E_s defined in the main text, while `raw_sq_sum` (raw second moment) and `abs_net_sum` (absolute first moment) serve as negative controls. These three statistics are evaluated on the full cross-product grid of drift intensity, workload, and probe parameters.

Parameter	Values
Score type	`dedrift_countscore` (primary), `raw_sq_sum` (negative control), `abs_net_sum` (negative control)
q	$\{0.2, 0.5, 0.9\}$
Workload	{bimodal, heavy_share, diffuse}
Base config	$L=4096$, $m_0=16$, $W=4$, $V=2$, $k=3$, $B=10$
Extended (V/W)	$V\in\{2,3\}$, $W\in\{2,4\}$, $B=10$
Extended (high B)	$V\in\{2,3\}$, $W\in\{2,4\}$, $B\in\{100,500\}$

Table S8: Experimental setup for drift robustness evaluation (full cross-product grid, 30 repetitions per point).

S3.1 Cross-Configuration Stability of Sizing

Table S9 expands selected rows from the main-text Table 3 with additional workloads. Because the full cross-product grid is too large to present in tabular form, Figure S5 summarizes the extended high-B settings ($B\in\{100,500\}$) compactly, plotting `p90(|rel_err|)` rather than signed error. The same separation pattern persists throughout the full grid: de-drifted energy remains stable and nearly invariant to drift; the raw second moment still exhibits failures on the order of 10^4% under strong drift; and the first moment consistently falls in the 20%–96% range. The V/W parameter sweep yields the same conclusion and is omitted for brevity.

After applying the Bessel-type scaling $\hat{D}^{\text{scaled}} = \hat{D}\cdot m_0/(m_0-1)$, the systematic underestimation converges from approximately -9% to approximately -3%. Table S10 summarizes the numerical ranges across the base configuration and two extended sweeps: the convergence magnitude remains at approximately 6 percentage points across all configurations, indicating that this correction effect is governed by

score_type	q	Workload	median(rel_err)	p90(rel_err)	Interpretation
raw_sq_sum	0.2	bimodal	+16659.2%	+17418.0%	Orders-of-magnitude bias (failure)
raw_sq_sum	0.9	bimodal	+29576.3%	+31180.0%	Orders-of-magnitude bias (failure)
raw_sq_sum	0.5	bimodal	+0.4%	+5.4%	Normal
raw_sq_sum	0.9	diffuse	+132.5%	+134.3%	Significant bias
dedrift_countscore	0.2	bimodal	-9.1%	-4.1%	Normal
dedrift_countscore	0.5	bimodal	-9.1%	-5.0%	Normal
dedrift_countscore	0.9	bimodal	-8.9%	-3.7%	Normal
dedrift_countscore	0.5	diffuse	-9.3%	-8.9%	Normal
dedrift_countscore	0.5	heavy_share	-9.2%	-6.6%	Normal
abs_net_sum	0.5	bimodal	-95.0%	-94.9%	Significant bias

Table S9: Representative numerical values from Table 3: sizing error under drift for different score types.

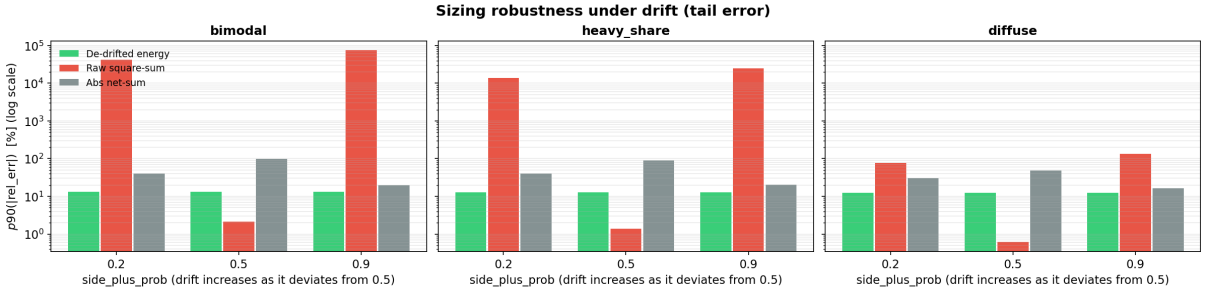


Figure S5: Drift robustness of sizing under extended configurations ($B \in \{100, 500\}$, $V \in \{2, 3\}$, $W \in \{2, 4\}$). The figure plots p90(|rel_err|) to compare error magnitude. The de-drifted statistic remains stable across the full drift range. Y-axis is on a log scale.

Metric	Base config (§5.3)	Extended (V/W)	Extended B ($\{100, 500\}$)
median(rel_err) (across drift range)	[-9.25, -9.17]	[-10.41, -10.28]	[-10.65, -10.46]
median(rel_err_scaled) (across drift range)	[-3.19, -3.12]	[-4.44, -4.30]	[-4.69, -4.49]
Convergence magnitude (pp)	$\approx$ +6.06	$\approx$ +5.98	$\approx$ +5.96

Table S10: Sizing bias before and after Bessel correction (units: %).

B	Score type	q=0.2	q=0.5	q=0.9
10	dedrift_countscore	0.0153	0.0166	0.0162
	abs_net_sum	0.0198	**0.425**	0.0187
500	dedrift_countscore	0.430	0.426	0.428
	abs_net_sum	0.490	**0.745**	0.411

Table S11: K_{eff}/L for two statistics under heavy_share (B=10 and B=500). De-drifted energy remains stable across both B levels and all three drift conditions; the first moment exhibits anomalous inflation at q=0.5.

the structural properties of the statistic.

S3.2 Cross-Configuration Stability of Gating

The main text (§5.3) confirmed that K_{eff}/L computed from de-drifted energy remains stable under drift in the base configuration (B=10). Table S11 and Figure S6 extend this validation to high-pressure configurations (B=500). The K_{eff}/L of de-drifted energy stays within the narrow band of 0.426–0.430 across three drift conditions, despite the absolute value increasing from ~ 0.016 (B=10, Table S11) to ~ 0.43 due to the larger B. As a control, the first moment (abs_net_sum) at q=0.5 is inflated to 0.745.

S3.3 Insensitivity to De-drifting

The preceding sections established the necessity of de-drifting for sizing and gating. A natural follow-up question is whether triage similarly requires de-drifting. Figure S7 compares the minimum cand_frac required by the three statistics on the same drift sweep grid, using the operating point framework (§5.4). Within the tested range, all three statistics exhibit comparable variation with drift, and no additional failure mode induced by drift is observed in the triage dimension.

Accordingly, we use the same de-drifted energy for sizing,

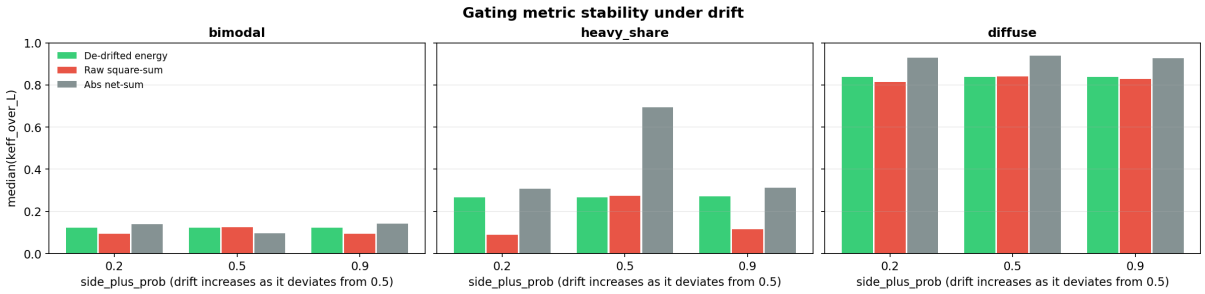


Figure S6: Drift stability of gating under high-pressure configurations ($B \in \{100, 500\}$). The K_{eff}/L of de-drifted energy remains nearly constant within each workload as drift varies; the first moment exhibits anomalous inflation at q=0.5.

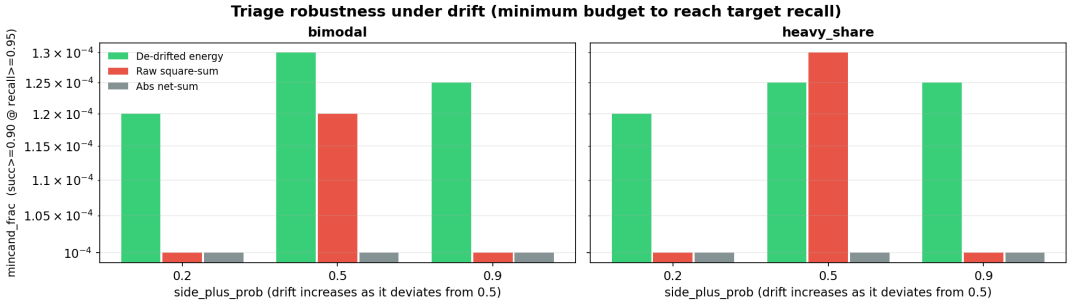


Figure S7: Triage under drift sweep. The figure plots the minimum `cand_frac` required by the three statistics as drift varies.

gating, and triage. This choice introduces no additional probe overhead and incurs no observable penalty on triage metrics.

S4 Sizing and Gating: Supplementary Derivations

This section provides supplementary derivations and empirical evidence for §5.3. The Sizing portion covers the theoretical derivation of scaling laws (§S4.1), sources and quantitative bounds of negative bias (§S4.2), and stability evidence for accuracy across a wider parameter range (§S4.3). The Gating portion covers metric selection and semantic validation (§S4.4), the scale separation of K_{eff}/L and theoretical analysis of the occupancy threshold (§S4.5), and the necessity argument for the three-state design (§S4.6).

S4.1 Fusion Law and Variance Scaling

This section supplements the derivations of two key phenomena: first, why multi-view fusion yields a $1/\sqrt{V}$ variance reduction; second, why the de-drift term eliminates the influence of q in expectation under drift conditions, along with an approximate variance scaling that aids understanding of the m_0/V scaling laws.

S4.1.1 Theoretical Derivation of the Fusion Law

The multi-view fusion law (Std $\propto 1/\sqrt{V}$) follows from the basic property of averaging independent estimators. Let the sizing estimates $\hat{D}_v$ from each view be independent and identically distributed with common variance $\text{Var}(\hat{D}_v) = \sigma^2$. The fused estimate is:

$$\hat{D} = \frac{1}{V}\sum_{v=1}^{V} \hat{D}_v.$$

By independence, the variance of the fused estimate is $\text{Var}(\hat{D}) = \sigma^2/V$, so $\text{Std}(\hat{D}) = \sigma/\sqrt{V} \propto V^{-0.5}$. In log-log coordinates, $\log(\text{Std}) = \text{const} - 0.5 \cdot \log(V)$, i.e., the slope is -0.5.

We now derive the expectation and variance of the de-drift statistic itself, to explain both drift invariance and accuracy scaling laws (§5.3).

Let $M = L \cdot m_0$ denote the total cell count of a single view. In implementation, each diff key updates only k distinct positions among the $W \cdot m_0$ candidate cells within the window of its corresponding segment; here we adopt a marginal distribution approximation where, after uniform hashing of window start positions on the section axis, these updates are treated as landing on k distinct positions among the global M cells, yielding an interpretable closed form (the main text §4.3 discusses the applicability and boundaries of this approximation). The symmetric difference contains D diff keys in total. Each diff key produces a sign $\sigma_i \in \{+1, -1\}$, where $\Pr(\sigma_i = +1) = q$. Let $X_{ij} \in \{0, 1\}$ indicate whether the i-th key hits the j-th cell;

then the cell count is:

$$C_j = \sum_{i=1}^{D} \sigma_i X_{ij}.$$

Define the global aggregate statistics:

$$S_{\text{net}} = \sum_{j=1}^{M} C_j, \qquad S_{\text{sq}} = \sum_{j=1}^{M} C_j^2.$$

Expanding the square and summing over j:

$$S_{\text{sq}} = \sum_j \sum_i \sigma_i^2 X_{ij} + \sum_j \sum_{i \neq \ell} \sigma_i \sigma_\ell X_{ij} X_{\ell j}.$$

In the first term, $\sigma_i^2 \equiv 1$, and each key hits k cells, so:

$$\mathbb{E}\left[\sum_j \sum_i \sigma_i^2 X_{ij}\right] = Dk.$$

In the second term, $\mathbb{E}[\sigma_i \sigma_\ell] = \mathbb{E}[\sigma_i]\mathbb{E}[\sigma_\ell] = (2q-1)^2$, and for two distinct keys, the expected size of the intersection of their k-element landing sets is approximately k^2/M (note: fixing the first key's k cells, the probability that any one cell is hit by the second key is approximately k/M; summing over the k cells gives expected intersection $\approx k \cdot (k/M)$), so:

$$\mathbb{E}\left[\sum_j \sum_{i \neq \ell} \sigma_i \sigma_\ell X_{ij} X_{\ell j}\right] \approx D(D-1)(2q-1)^2 \cdot \frac{k^2}{M}.$$

On the other hand, note that $S_{\text{net}} = \sum_j C_j = \sum_i \sigma_i \sum_j X_{ij} = k \sum_i \sigma_i$, so:

$$\mathbb{E}[S_{\text{net}}^2] = k^2 \cdot \mathbb{E}\left[\left(\sum_i \sigma_i\right)^2\right] = k^2 \left(D + D(D-1)(2q-1)^2\right).$$

Therefore:

$$\begin{aligned}\mathbb{E}\left[S_{\text{sq}} - \frac{S_{\text{net}}^2}{M}\right] &\approx Dk + D(D-1)(2q-1)^2 \frac{k^2}{M} \\ &\quad - \frac{k^2}{M}\left(D + D(D-1)(2q-1)^2\right) \\ &= Dk\left(1 - \frac{k}{M}\right).\end{aligned}$$

This result has two implications. First, the $(2q-1)^2$ terms cancel exactly after de-drifting, so **the expectation does not depend on** q. This is why the de-drifted statistic remains stable under side imbalance in the main-text experiments (§5.3). Second, $1 - k/M$ reflects the energy dilution from cross-key collisions in the finite cell space, consistent with the $(1 - k/M)$ term in the sizing denominator used in implementation.

This derivation uses the global quantity $S_{\text{sq}} - S_{\text{net}}^2/M$, whereas the experiments use the per-section form $\sum_s E_s$. The two differ by a small constant-factor term that diminishes as m_0 increases, but the core conclusion—that the expectation does not depend on q—holds for both definitions.

Bessel Correction (Degrees-of-Freedom Correction) In implementation, $E_s = \sum_{j=1}^{m_0} (\Delta_j - \bar{\Delta}_s)^2$ is equivalent to the unnormalized sample variance of the m_0 cell counts within section s. By the degrees-of-freedom identity,

$$\mathbb{E}[E_s] = (m_0 - 1) \cdot \text{Var}(\Delta_j \mid \text{section } s),$$

so when the target quantity is written at the scale of $m_0 \text{Var}(\cdot)$, the estimator based on $\sum_s E_s$ carries a first-order multiplicative bias of $(m_0 - 1)/m_0$. Multiplying the sizing output by $m_0/(m_0 - 1)$ eliminates this bias—this is precisely the Bessel correction factor used in the main text and evaluation.

S4.1.2 Zero Energy and Skip Signal

By definition, $E_s = \sum_{j=1}^{m_0} (\Delta_j - \bar{\Delta}_s)^2 \geq 0$, with $E_s = 0$ if and only if all cell difference counts Δ_j within section s are identical. If $\text{net_sum}_s = \sum_j \Delta_j = 0$ also holds, then $\Delta_j \equiv 0$, so section s carries zero differential observation.

This section-level fact lifts to a segment-level skip criterion. For a segment p, let $W_v(p)$ denote its mapped width-W section window in view v. If all sections in $W_v(p)$ satisfy both $E_s^{(v)}=0$ and $\text{net_sum}_s^{(v)}=0$ for any single view v, then every probe cell reachable from segment p in that view is zero. Because any diff key belonging to p would update $k \geq 1$ cells within $W_v(p)$, the zero observation implies that no such key exists—or that its contributions are exactly canceled by collisions from other segments, an event whose probability is negligible under random hashing and non-adversarial inputs.

The segment can therefore be excluded from downstream verification. Requiring the condition to hold across all V views provides a strictly stronger guarantee.

S4.1.3 Variance Scaling and Poisson Approximation

To provide an interpretable scaling, we use a common Poisson approximation: treating the load on each cell as approximately independent and with $\mathbb{E}[C_j^2]$ at the scale determined by $\lambda = Dk/M$. Under this approximation, $S_{\text{sq}} - S_{\text{net}}^2/M$ is approximately the sum of squares of M nearly independent terms, whose leading variance term (analogous to the $\text{Var} = 2n$ property of the χ^2 distribution) can be written as:

$$\text{Var}\left(S_{\text{sq}} - \frac{S_{\text{net}}^2}{M}\right) \approx 2M\lambda^2 = 2 \cdot \frac{(Dk)^2}{M}.$$

The mean is $\mathbb{E}[\cdot] \approx Dk(1 - k/M) \approx Dk$, so the relative standard deviation under a single view satisfies:

$$\text{Std}(\text{rel_err}) = \frac{\text{Std}(\hat{D})}{D} \approx \Theta\left(\sqrt{\frac{1}{M}}\right) = \Theta\left(\sqrt{\frac{1}{L \cdot m_0}}\right),$$

and after independently averaging over V views, further yields the $1/\sqrt{V}$ scaling:

$$\text{Std}(\text{rel_err}_{\text{fusion}}) \approx \Theta\left(\sqrt{\frac{1}{V \cdot L \cdot m_0}}\right).$$

This approximation accounts for three trends observed in the sizing subsection (§5.3): increasing V provides a predictable $1/\sqrt{V}$ variance reduction; increasing m_0 reduces error but with diminishing returns; and relative error depends only weakly on D (the leading term in D cancels).

The Poisson approximation is used to explain scaling slopes: with typical parameters ($V = 4, L = 2048, m_0 = 32$), the approximation gives Std(rel_err) of about 0.28%, while the experimental observation is about 2.25%; however, the scaling slopes in V and m_0 remain consistent.

S4.2 Locality Bias of Sizing

The sizing estimator $\hat{D} = \sum_s E_s/[k(1-k/M)]$ in §4.3 uses the global pool $M = L \cdot m_0$ in its denominator. This is the marginal distribution approximation obtained by treating window start positions as uniformly hashed on the section axis: when differing keys are distributed across multiple segments, cross-window pairings dominate, and the approximation is reasonable. However, when differences are highly concentrated in a few segments, each key's hash landing points are strictly confined to the W sections (a total of $M_{\text{local}} = W \cdot m_0$ cells) within its segment's window, and the effective collision pool shrinks to the local scale. In this case, the estimator's expectation satisfies

$$\mathbb{E}[\hat{D}] \approx D \cdot \frac{1-k/M_{\text{local}}}{1-k/M} < D,$$

i.e., the global approximation $(1-k/M)$ underestimates the intensity of local collisions, causing systematic underestimation. The magnitude of the bias is determined by the gap between M_{local} and M: with typical parameters (k=2, W=4, m_0=32, L=2048), M_{local}=128, M=65,536, and the formula above gives a negative bias upper bound of approximately 1.6%. This bias decreases as W/L shrinks—when differences are distributed across multiple segments or L is large, the effective pool approaches the global M, and the bias approaches zero. The main conclusion from §S4.1—that the expectation does not depend on q—is unaffected by this approximation.

In experiments, one can observe that the negative bias is more pronounced when m_0 is small (because k/M_{local} is larger), and converges rapidly as m_0 increases (Figure S9). If explicit compensation is desired, M_{local} can replace M in the denominator; however, at typical deployment parameters, the 1.6% bias magnitude usually does not have operational impact.

S4.3 Supplementary Empirical Evidence for Sizing Accuracy

Baseline evidence under the standard setting For sizing, the principal accuracy metric in the main text is Std(rel_err). This section provides the additional per-setting statistics, broader parameter sweeps, and supporting evidence underlying the sizing results. The main text §5.3 established three scaling laws: the fusion law (Std $\propto V^{-0.5}$), the accuracy-memory trade-off of m_0, and the weak dependence of relative error on discrepancy size D. We first revisit the fusion law under the baseline configuration. Across the parameter grid summarized in Figure S8, log-log regression of Std(rel_err) against V yields a mean slope of -0.498 (std. 0.050), close to the theoretical value -0.5; Table S12 lists representative values showing the corresponding decrease in relative error standard deviation as V increases. The third law is an implicit prerequisite for the first two—if relative error drifted significantly with D, the promise of predictable accuracy from a fixed memory budget would no longer hold. This section therefore verifies the robustness of all three laws under two parameter settings.

V	1	2	4	8	16
`Std(rel_err)` (%)	3.53	2.66	2.25	1.33	1.01

Table S12: Fusion law under a representative configuration (m_0=32, L=2048).

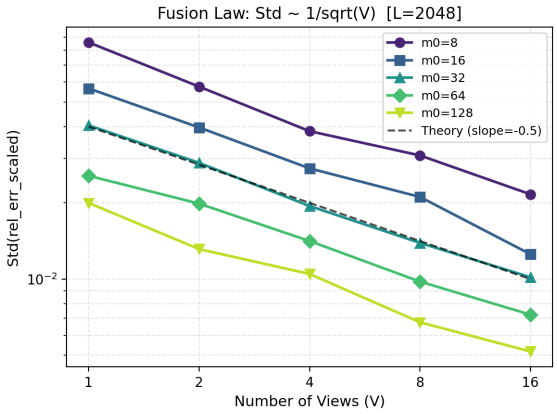


Figure S8: Empirical verification of the fusion law: across a parameter grid covering 90 (m_0, L, D) configurations, the mean log-log slope is -0.498, in close agreement with the theoretical prediction of -0.5.

Robustness under a more demanding setting Under the baseline configuration ($V \in \{1,2,4,8,16\}$, $m_0 \in \{8,\ldots,128\}$, $L \in \{1024,\ldots,4096\}$), as D spans from 10^2 to 10^7, the `Std(rel_err)` aggregated across all (V, m_0, L) fluctuates around 2%, indicating that relative accuracy is primarily governed by probe capacity parameters rather than discrepancy magnitude (Figure S10). Additionally, we repeated the verification under a more demanding system and fault configuration: larger scale P=10^6, higher fault density B=100, stronger

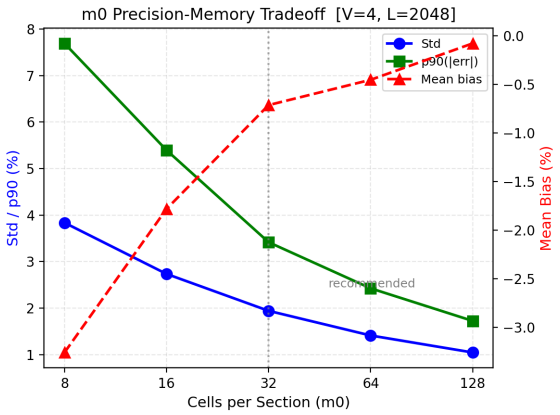


Figure S9: Accuracy-memory trade-off of m_0: Std(rel_err) decreases monotonically with m_0, with diminishing returns beyond $m_0 = 32$; the negative bias converges simultaneously.

asymmetry ($q = 0.1$), and lighter updates (W=2, k=1). Under this stress setting, all three laws are reproduced. For the fusion law, log-log regression over (L, m_0, D) combinations for $V \in \{1, 4, 16\}$ yields a mean slope of approximately **-0.489** (median **-0.494**), consistent with the theoretical -0.5 (Figure S11). The accuracy-memory trade-off of m_0 likewise holds: the aggregated `Std(rel_err_scaled)` decreases monotonically with m_0 and exhibits diminishing returns (Figure S12). And the weak dependence on D continues to hold over a larger span: as D ranges from 10^3 to 10^8, no order-of-magnitude deterioration in relative error is observed.

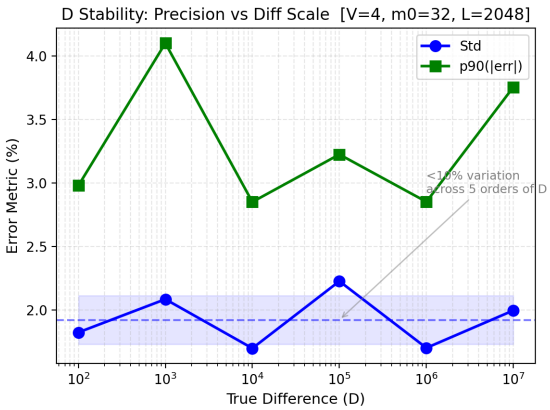


Figure S10: Weak dependence of sizing relative accuracy on D: as D varies across multiple orders of magnitude, the relative error metrics show only modest fluctuation overall.

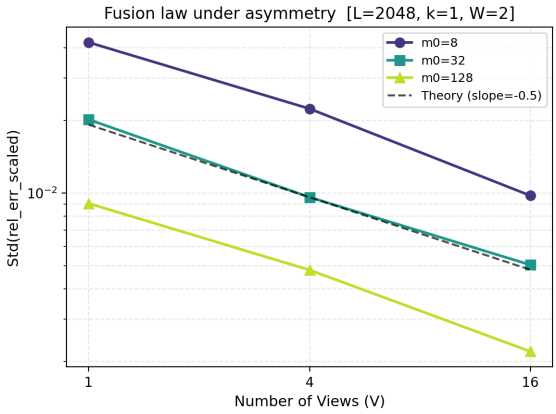


Figure S11: Reproduction of the fusion law under a more demanding setting: mean log-log slope approximately -0.489, consistent with the theoretical -0.5.

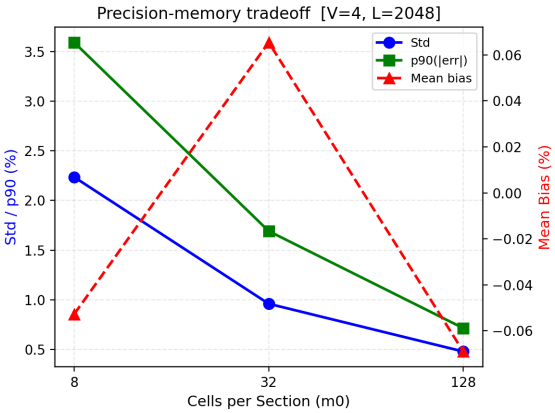


Figure S12: Reproduction of the m_0 accuracy-memory trade-off under a more demanding setting. Unlike the baseline configuration (Figure S9), after Bessel correction the mean bias has dropped to within 0.06% and fluctuates around zero.

S4.4 Metric Selection and Semantic Validity for Gating

Separability of candidate metrics Gating requires a scalar metric to quantify the concentration of the energy spectrum (§4.4). Before settling on K_{eff}/L, we compared four candidate statistics across a sweep covering bimodal ($B \in \{1,5,10,50,200\}$), diffuse ($D = 10^5$), and heavy_share workloads over multiple (P,L) configurations (with V=2, m_0=16, W=4, k=2; 50 repetitions per group): K_{eff}/L, Top-1/32 energy share, Gini coefficient, and normalized entropy. As shown in Figure S13, K_{eff}/L exhibits the largest separation between bimodal and diffuse (medians differ by approximately 36×), and this separation remains stable across different (P,L) configurations. Furthermore, K_{eff}/L, based on the continuous definition of the Herfindahl index, serves as a continuous measure of spatial concentration without relying on discrete binning or top-k truncation.

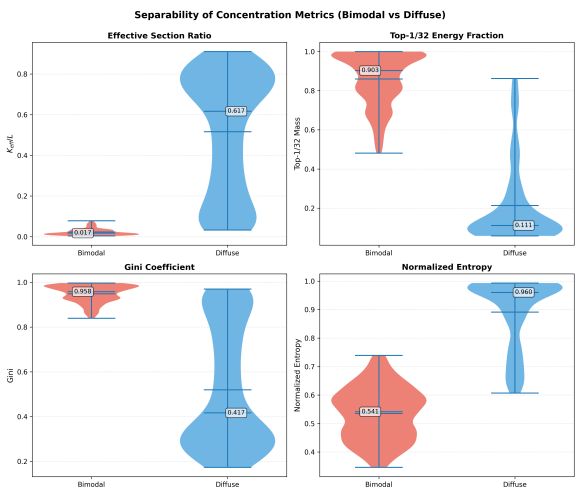


Figure S13: Comparison of gating metric candidates: K_{eff}/L achieves the largest separation (approximately 36×) with the smallest distributional overlap.

Semantic validity with respect to spatial extent Beyond separation, a concentration metric should primarily reflect how dispersed the differences are across segments, rather than being dominated by the discrepancy size D. To verify this property, we compared two bimodal settings: one holding total D fixed and diluting each segment's differences as B increases, and another holding per-segment difference d_{big}=10000 fixed so that total D grows with B. Taking a representative slice at L=2048 (Table S13), the response of K_{eff}/L to B is nearly identical under the two settings, indicating that the statistic's magnitude is primarily driven by B (spatial extent) rather than directly by total scale D.

B	Fixed D (dilution) setting	Fixed d_{big} ($D = B \times 10^4$) setting
1	0.0018	0.0018
10	0.017	0.017
50	0.078	0.079
200	0.25	0.25

Table S13: Response of K_{eff}/L to B: fixed D vs. fixed d_{big}.

S4.5 Scale Separation and Observability Boundary

This section further provides a theoretical derivation of the scale separation between the two discrepancy classes, and then shows through empirical evidence how this separation degrades under low occupancy. Together, these results establish the basis for the occupancy threshold used by gating.

Scale separation for concentrated faults Let $w_s = E_s/\sum_s E_s$, $H = \sum_s w_s^2$, $K_{\text{eff}} = 1/H$. Suppose there exists a hot section set $\mathcal{S}$ ($|\mathcal{S}| = S$) whose energy mass share is $\sum_{s\in\mathcal{S}} w_s = 1-\varepsilon$. This paragraph expands the derivation details of $K_{\text{eff}} \le S/(1-\varepsilon)^2$ referenced in §5.3. Note that:

$$H = \sum_s w_s^2 \ge \sum_{s\in\mathcal{S}} w_s^2 \ge \frac{(\sum_{s\in\mathcal{S}} w_s)^2}{|\mathcal{S}|} = \frac{(1-\varepsilon)^2}{S},$$

and therefore:

$$K_{\text{eff}} = \frac{1}{\sum_s w_s^2} \le \frac{S}{(1-\varepsilon)^2}.$$

This inequality provides an upper bound. When the energy within the hot set is approximately uniformly distributed across sections (i.e., ε is small and each hot section's weight is of the same order), the bound becomes tight and K_{eff} falls at the scale of S, so:

$$\frac{K_{\text{eff}}}{L} = \Theta\left(\frac{S}{L}\right).$$

For concentrated faults, B dirty segments each cover W contiguous positions on the section axis. When $B \cdot W \ll L$, the windows of different dirty segments are nearly non-overlapping on the section axis, and the total number of hot sections can be approximated as $S \approx B \cdot W$, giving K_{eff}/L at the scale of $B \cdot W/L$. With typical parameters $L = 2048$, $B = 10$, $W = 4$, the theoretical estimate is $K_{\text{eff}}/L \approx BW/L = 40/2048 \approx 0.020$; the corresponding experimental median for bimodal in §5.3 is approximately 0.017, consistent in order of magnitude. Note that when $B \cdot W$ is no longer much smaller than L, window overlap causes the effective number of hot sections to fall below $B \cdot W$, and the above approximation breaks down—this is precisely the microscopic mechanism behind the capacity-axis degradation in §5.4.

Diffuse-case scaling and observability boundary For diffuse differences, energy is approximately uniformly spread across the touched effective sections. Let the number of touched sections be $L_{\text{eff}} = \text{occupancy} \cdot L$, with each section's energy weight approximately equal, $w_s \approx 1/L_{\text{eff}}$. Then $H = L_{\text{eff}} \cdot (1/L_{\text{eff}})^2 = 1/L_{\text{eff}}$, so $K_{\text{eff}} = L_{\text{eff}}$, i.e.:

$$\frac{K_{\text{eff}}}{L} \approx \frac{L_{\text{eff}}}{L} \approx \text{occupancy}.$$

When the occupancy of the diffuse case drops to the same order as $B \cdot W/L$, the diffuse-case K_{eff}/L is compressed into the same numerical interval as the concentrated case, and threshold discrimination becomes unreliable. This marks the observability boundary of the probe: below this region, the signal no longer supports reliable shape classification, so low-occupancy samples should be excluded from such decisions.

Empirical manifestation of the observability boundary We next examine representative configurations and a broader parameter sweep to show how the diffuse-case collapse appears in practice as occupancy decreases.

P	L	occupancy	K_{eff}/L
10^5	2048	1.000	0.877
10^4	4096	1.000	0.838
3×10^3	8192	0.367	0.321
10^3	4096	0.245	0.215
10^3	16384	0.061	0.054
10^3	32768	0.031	0.027

Table S14: Occupancy and K_{eff}/L for diffuse workloads under representative configurations. From top to bottom, occupancy decreases and K_{eff}/L systematically declines, consistent with the theoretical prediction $K_{\text{eff}}/L \approx$ occupancy.

Table S14 shows a set of representative configurations covering high, medium, and low occupancy levels. In the high-occupancy region (P=10^5 or P=10^4, occupancy reaching 1.0), K_{eff}/L stabilizes near 0.84–0.88, close to the theoretical upper limit. As occupancy decreases due to increasing L or decreasing P, K_{eff}/L systematically declines: when occupancy drops to 0.061, K_{eff}/L is only 0.054, already entering the numerical range that can overlap with borderline bimodal cases.

Figure S14 further shows the continuous trend across multiple (P,L) configurations: as occupancy decreases, K_{eff}/L declines approximately linearly, consistent with the theoretical prediction $K_{\text{eff}}/L \approx$ occupancy, indicating that this is a stable structural regularity.

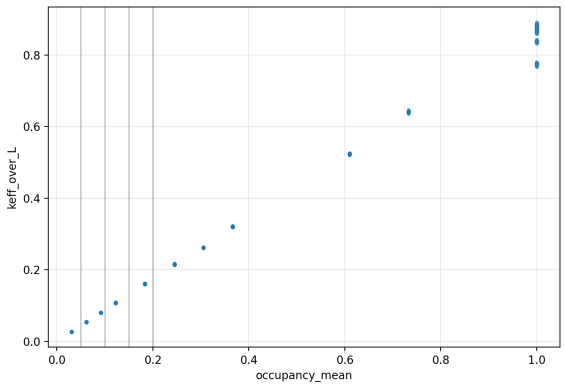


Figure S14: Relationship between K_{eff}/L and occupancy for diffuse workloads. Each point corresponds to the aggregated result of a (P,L) configuration group; on the right (high occupancy) K_{eff}/L approaches the theoretical upper limit, while on the left it systematically decreases with declining occupancy.

S4.6 Abstention Cost and Conditional Discrimination Quality

Why unconditional discrimination has a floor Figure 4 in the main text already showed the unconditional FAR–FRR trade-off, where FRR has a floor determined by low-occupancy samples. This section explains this phenomenon from two angles.

When τ_{occ}=0 (i.e., no occupancy threshold is set), all samples are forced into either bimodal or diffuse classification, and FAR and FRR cannot be simultaneously suppressed (see the corresponding curve in Figure 4). This degradation has two layers of causes. The first layer is statistical distortion: in low-occupancy samples, a large number of sections are never touched, and the diffuse-case K_{eff}/L undergoes the structural collapse analyzed in the previous section, overlapping with the bimodal value range—this is precisely the problem that the occupancy threshold is designed to solve, and excluding these samples from classification eliminates it.

The second layer is the inherent limit of the gating task itself: taking L=2048 as an example, B=200 corresponds to a dirty segment fraction of approximately 10%, at which point the pattern has already substantially diffused and exhibits genuine overlap with the diffuse class. Such overlap stems from the physical convergence of the two regimes in this parameter range and cannot be eliminated by raising τ_{occ}; it constitutes an unavoidable ambiguity zone for gating.

Abstention cost under the occupancy threshold The three-state design excludes the region with insufficient occupancy from classification, trading abstention for improved discrimination quality. Table S15 quantifies this trade-off: taking τ_{occ}=0.10 as an example, approximately 87% of diffuse samples still fall within the observable region, while the

τ_{occ}	diffuse classified	diffuse `unknown`	bimodal classified	bimodal `unknown`
0.05	95.7%	4.3%	25.0%	75.0%
0.10	87.0%	13.0%	8.3%	91.7%
0.15	82.6%	17.4%	8.3%	91.7%
0.20	78.3%	21.7%	8.3%	91.7%

Table S15: Effect of different τ_{occ} on the proportion of definitive classifications vs. `unknown`. The proportion of definitively classified samples is defined as $\Pr(\text{occupancy} \geq \tau_{occ})$; the remaining samples output `unknown`, for which the probe withholds a shape-based recommendation and deployment policy determines the downstream action.

majority of bimodal samples (especially those with small B) are excluded due to insufficient occupancy. This is operationally acceptable: low occupancy corresponds precisely to the concentrated pattern where triage is most effective, so an `unknown` verdict in this region withholds a shape recommendation rather than masking a diffuse fault. Deployment policy may choose to attempt a budget-capped triage pass informed by $\hat{D}$ and occupancy, but this decision is not driven by a shape verdict.

Conditional discrimination after filtering After the occupancy threshold filters out low-observability samples, the system computes conditional error rates only on definitively classified samples. Compared with the unconditional view in the main text (Figure 4), the conditional error rates can simultaneously approach 0 (Figure S15).

Because the workload configurations form a discrete grid, each $(\tau_{occ}, \tau_{gating})$ pair yields a single (FAR, FRR) point rather than a continuous curve. Dominated points—those for which another threshold setting achieves both lower FAR and lower FRR—carry no decision-relevant information and are omitted; the retained non-dominated frontier (Pareto front) is a standard presentation for discrete multi-objective trade-offs. Dashed lines serve as visual guides only.

S5 Operating Limits: Extended Scans and Deployment Details

This section provides supplementary evidence and derivations for §5.4. We first present slice views and extended parameter sweeps along P/L and L/B (§S5.1), then provide the theoretical derivation of the expected candidate-set size and the V–L allocation under a fixed budget (§S5.2), and finally give example verification of the runtime stopping rule (§S5.3).

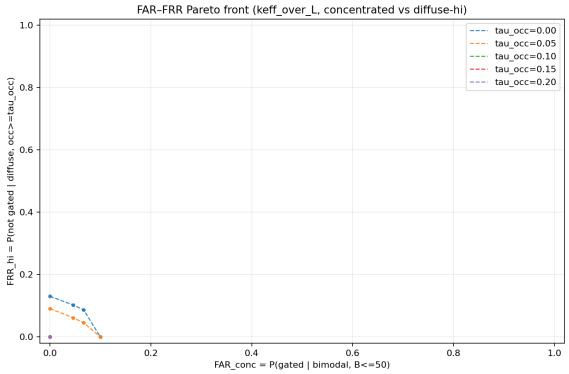


Figure S15: Conditional FAR_{cond}–FRR_{cond} trade-off on definitively classified samples (read in the same way as Figure 4). FAR is computed only on concentrated workloads with $B \leq 50$ (excluding B=200, whose pattern has already converged toward diffuse); FRR is computed only on diffuse samples with occupancy $\geq \tau_{occ}$. When $\tau_{occ} \geq 0.10$, both error rates simultaneously approach 0.

S5.1 Supplementary Evidence Along P/L and L/B

This subsection supplements the degradation-axis slices and extended parameter sweeps from §5.4.

S5.1.1 L/B Slices and Extended Parameter Range

Figure S16 presents the basic trend along L/B in slices at fixed P/L: the operating-point candidate count `cand_cnt` decreases monotonically with increasing L/B across all configurations, and larger V yields lower absolute cost at the same L/B.

Figure S17 extends the parameter range to larger values of L. The candidate count `cand_cnt` maintains an overall decreasing trend with increasing L/B, and the suppression effect of V remains consistent under the extended parameters. For a few panels with W=2 and V=2, a plateau or step-like pattern can be observed at the high-L/B end; this phenomenon is related to the discrete selection of operating points: when windows are narrow and the number of views is small, ranking stability is more sensitive, and the discrete grid of `top_r` values in the experiment causes the minimum feasible budget to jump between adjacent bins, producing locally non-smooth curves. The curves become smoother with larger V or larger W, and the overall trend is unchanged.

S5.1.2 P/L: Extended Parameter Range

Figure S18 extends the P/L analysis to higher P/L tiers. The candidate count increases monotonically with P/L, and the suppression effect of V remains consistent at higher tiers. The tail-risk metric `halo` (p90) follows exactly the same trend: in

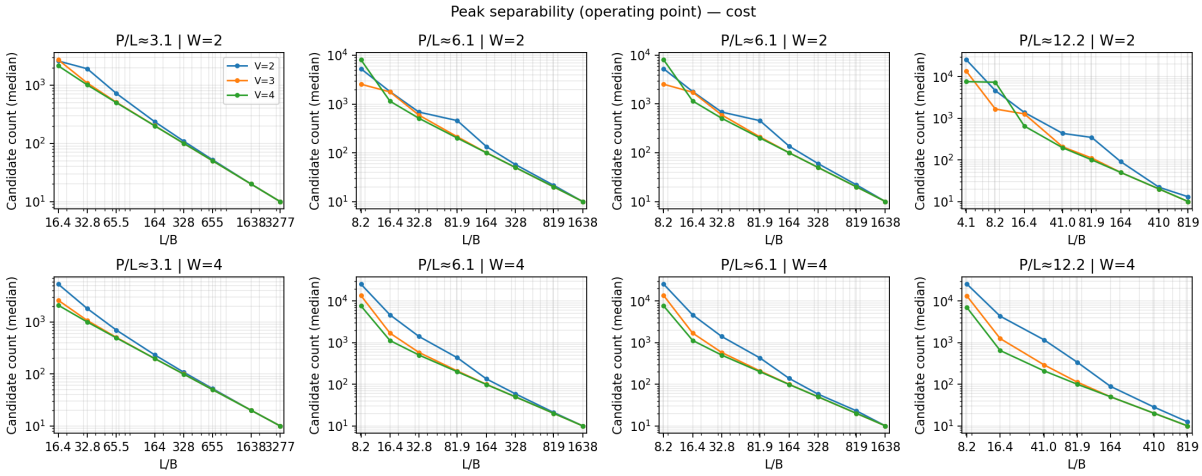


Figure S16: L/B slices at fixed P/L, operating-point `cand_cnt` as a function of L/B. The three curves in each panel correspond to V=2, 3, 4 (blue, orange, and green, respectively). Candidate cost decreases monotonically with increasing L/B across all configurations; larger V yields lower absolute cost at the same L/B.

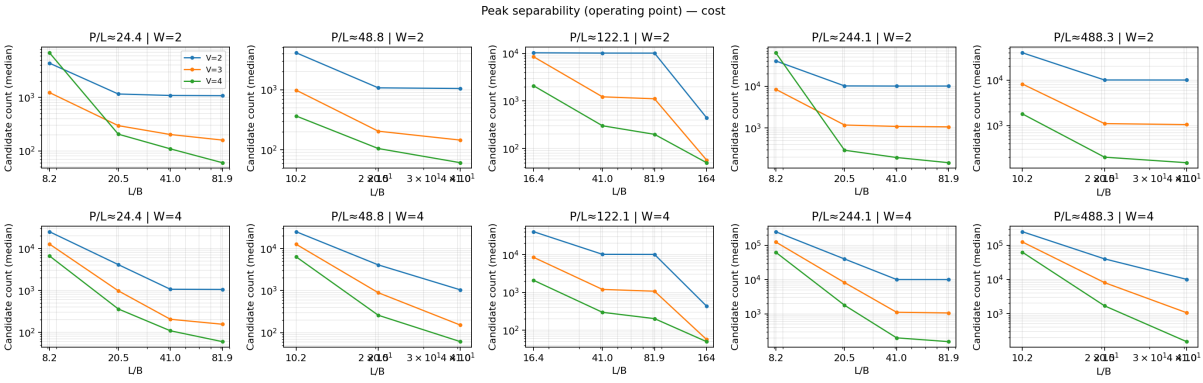


Figure S17: Operating-point `cand_cnt` (median) as a function of L/B under the extended parameter range. Configuration is the same as the base-range version, but with a wider range of L values. Log scale.

this setting it differs from `cand_cnt` only by a constant factor on the order of P/B, so it is not plotted separately.

P	P/L	V	`cand_cnt` (med)	`halo` (p90)
10,000	1.2	2	127	1.35
10,000,000	1221	2	25,044	251.87
10,000,000	1221	4	163	1.69

Table S16: Scale amplification and the suppression effect of V (L=8192, W=2, B=10, `top_r`=0.05).

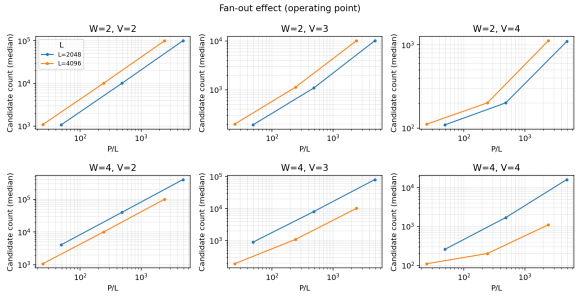


Figure S18: `cand_cnt` (median) as a function of P/L under the extended parameter range. Log scale.

S5.2 Theoretical Analysis of Candidate-Set Size and Capacity Threshold Under a Fixed Budget

This subsection studies how a fixed digest budget should be allocated between the number of views V and the section-axis length L. Building on the candidate-set framework of Equations (1) and (2) in the main text, and fixing m_0=32 as a practical balance between sizing accuracy and memory cost for digest sizing (Supplement S4.3, especially Figure S9), we derive the fixed-budget trade-off between V and L, the relaxed-model threshold for when increasing V remains beneficial, and the associated capacity threshold. We then compare these threshold predictions against numerical results. The focus here is offline allocation across cycles; the runtime adaptation of the truncation budget through `top_r` is separately in §S5.3.

S5.2.1 Derivation of the V–L Trade-off and Capacity Thresholds

Within this framework, the candidate set consists of B truly dirty segments and clean segments that are falsely included due to window overlap, whose expected count is $(P-B)\,p_{\text{halo}}^{V}$. The following derivation assumes that L/B is sufficiently large so that the high-energy windows of the B dirty segments are well-dispersed on the section axis and do not compete for top-ranked slots; the number of top windows t required to cover all dirty segments is then of the same order as BW, written as $t \approx \alpha BW$, where $\alpha \geq 1$ is a mild inflation factor accounting for ranking redundancy that is approximately 1 within the parameter range of this paper. The failure boundary of this assumption is discussed below. Here B denotes the number of dirty segments at segment granularity, but in the relaxed model it is interpreted as an occupancy-scale quantity: it captures how many faults contribute distinct support on the section axis, while weak or highly dispersed discrepancies are absorbed into the effective redundancy factor α rather than inflating B itself. When heavy-tail noise compresses the ranking margin (§5.5), the effective α can be significantly larger than 1. Since $c \propto \alpha$ in Equation (S3), an increase in α linearly raises the per-view false-inclusion probability and correspondingly shrinks $V^* = 1/(ec)$, thereby proportionally tightening the capacity thresholds of Equations (S6) (e.g., the threshold becomes $L/B \gtrsim e\alpha W(2W-1)$). The closed-form structure itself is unchanged; what changes is the numerical threshold and the scale of the feasible region.

Structure under the budget constraint SCOUT's digest budget satisfies

$$M_{\text{bytes}} = 4\,V\,L\,m_0,$$

so $L = M_{\text{bytes}}/(4Vm_0)$. Substituting this constraint together with $t \approx \alpha BW$ into Equation (2) yields

$$p_{\text{halo}}(V) = \min\left\{1,\ \frac{\alpha BW(2W-1)}{L}\right\} = \min\{1,\ cV\}, \quad \text{(S2)}$$

where

$$c \triangleq \frac{4\alpha BW(2W-1)\,m_0}{M_{\text{bytes}}} \quad \text{(S3)}$$

is a dimensionless capacity parameter independent of V and L. Equation (S2) reveals the core trade-off under a fixed budget: increasing V provides exponential suppression through the intersection term p_{halo}^{V}, but simultaneously raises the per-view $p_{\text{halo}} = cV$ linearly because the fixed budget forces $L \propto 1/V$. The dominant factor in the false-inclusion term, $(\min\{1, cV\})^V$, is determined by the interplay of these two forces.

Threshold for beneficial increases in V When $cV \geq 1$, p_{halo} saturates to 1 and the intersection mechanism fails completely—regardless of how large V is, every clean segment is included in the candidate set. Therefore, the meaningful optimization interval is $cV < 1$ (the non-saturated regime). In this regime the min operator can be dropped, and the false-inclusion factor simplifies to $f(V) = (cV)^V$. Taking the logarithm and differentiating:

$$\ln f(V) = V\ln(cV), \qquad \frac{d\ln f}{dV} = \ln(cV) + 1.$$

Setting the derivative to zero gives the stationary point under continuous relaxation: $V^* = \frac{1}{ec}$. Since V is a positive integer

and the intersection mechanism requires at least two views, the practical condition is $V^* \geq 2$, equivalently

$$c \leq 1/(2e). \tag{S4}$$

Although B is generally unknown before diagnosis, this threshold provides a useful provisioning check: given a provisioned budget M_{bytes}, a conservative estimate $\hat{B}$, and a conservative choice of α, evaluating c from Equation (S3) and verifying $V^* \geq 2$ indicates that the budget is large enough, under the relaxed model, for additional views beyond $V=2$ to remain potentially beneficial. Substituting back into f yields the minimum false-inclusion factor predicted by the relaxed model at the continuous optimum:

$$f(V^*) = \left(\frac{1}{e}\right)^{1/(ec)} = \exp\left(-\frac{M_{\text{bytes}}}{4e\,\alpha B W(2W-1)\,m_0}\right). \tag{S5}$$

Equation (S5) expresses how low the relaxed-model prediction can be driven under a fixed total budget as an exponential decay in M_{bytes}/B. As B increases (c increases), the exponent shrinks and the predicted minimum rises; when $c > 1/(2e)$, $V^* < 2$, indicating that under the relaxed model the budget no longer supports beneficial increases beyond the minimum feasible configuration $V=2$.

Feasibility and capacity thresholds Substituting back the original parameters, $cV = \alpha BW(2W-1)/L$, so the non-saturation condition $cV < 1$ is equivalent to $L/B > \alpha W(2W-1)$. This is a necessary but weak condition: it merely prevents p_{halo} from saturating to 1, but does not guarantee that increasing V is beneficial. The practically useful bound is the *capacity threshold*, which requires $V^* \geq 2$ (Equation S4) and is a factor of e more stringent:

$$\frac{L}{B} \gtrsim e\,\alpha W(2W-1). \tag{S6}$$

Geometrically, this states that the section axis must be long enough to accommodate the coverage footprints and halo neighborhoods of B concurrent faults without driving the per-view halo probability into saturation. Since L is determined by the allocation $L = M_{\text{bytes}}/(4Vm_0)$ rather than chosen independently, the same threshold also serves as a provisioning rule. At the minimum multi-view configuration $V=2$, the budget should satisfy

$$M_{\text{bytes}} \gtrsim 8e\,\alpha B W(2W-1)\,m_0.$$

Equivalently, for a fixed budget, the same relaxed threshold implies a planning capacity

$$B_{\max} = \frac{M_{\text{bytes}}}{8e\,\alpha W(2W-1)\,m_0}.$$

For example, with $W=2$, $\alpha=1$, and $m_0=32$, the threshold gives $L/B \gtrsim 6e \approx 16.3$, and a 512 KiB digest budget corresponds to $B_{\max} \approx 126$ under the relaxed threshold. The transition band in the main-text heatmap (Figure 5) appears at $L/B \sim$ 10–50, consistent with this order of magnitude.

Role of P/L The above thresholds are entirely determined by L/B and W, while P appears in the derivation only in the coefficient $(P-B)$, seemingly amplifying the absolute candidate count only linearly. This conclusion relies on the assumption that $t_{\text{op}} \approx \alpha BW$ does not vary with P/L—under a pure bimodal workload (dirty segments have strong signals and the rest are completely clean), the section energy ranking is determined by the dirty-segment signals, so $\alpha \approx 1$ is stable.

When the workload contains significant background noise (e.g., the mixed workloads of §5.5), a higher P/L implies severe hash collisions. A large number of clean segments (as well as slightly dirty ones) spread background energy across the section axis, interfering with the ranking of genuine faults. To ensure that genuinely large faults are not displaced from the candidate list, the system must relax the truncation quota (i.e., the redundancy factor α mentioned above becomes greater than 1).

However, this degradation induced by P/L is mild and gradual: it merely increases the difficulty of ranking, forcing the selection of a few more candidate windows. This is fundamentally different from an increase in the number of concurrent faults B—an increase in B physically fills the remaining space on the section axis ($c \propto B$), which saturates the section axis more directly.

S5.2.2 Numerical Verification Under a Fixed Budget

Building on Equation (2) in §4.2, we note that the derivation assumes independently scattered window start positions and therefore yields a conservative relaxed approximation. In practice, the top-ranked high-energy windows tend to cluster tightly around the true faults; the geometric overlap makes the effective halo neighborhood much smaller than the estimate under the independence assumption, so the actual false-inclusion probability is typically lower than this approximation. We next test whether this approximation still provides useful guidance under a fixed budget.

Prediction accuracy of the benefit threshold Given a fixed probe budget, increasing V means decreasing L. The continuous relaxation above yields the stationary point $V^* = 1/(ec)$ and the corresponding benefit threshold $p_{\text{halo}} = 1/e$. We now test the discriminative power of this threshold under discrete parameters. From the operating-point data of §5.4, we pair configurations with the same total budget ($V \cdot L$ fixed) at $V=2$ and $V=4$, yielding 55 pairs. For each pair, we compute $p_{\text{halo}} = \min\{1,\, t_{\text{op}}(2W-1)/L\}$ at the $V=2$ operating point and use $p_{\text{halo}} < 1/e$ as the prediction rule for "increasing V is beneficial," then check whether this agrees with the actual change in candidate cost. After excluding pairs where candidate costs are essentially tied, the rule achieves a prediction accuracy of approximately 90% among the 41 effective pairs. In terms of absolute values, the formula overestimates by roughly $2\times$ at low P/L and more significantly at high P/L

due to clustering effects; however, the discriminative power of the threshold $p_{\text{halo}} = 1/e$ is unaffected, indicating that the scaling laws derived from continuous relaxation remain valid under discrete parameters.

Reversal cases and the asymmetry between L/B and P/L More informative are the reversal cases—where increasing V while decreasing L worsens candidate cost. Table S17 shows the most prominent example: with B=1000 and $V \cdot L$=32768, $p_{\text{halo}} \approx 0.30$ at V=2 already approaches the benefit threshold. Increasing V to 4 halves L to 8192, saturating p_{halo} to 1.0 and forcing the operating point to push `top_r` to 0.5, inflating candidate cost by roughly 4×. This failure path—V increases $\rightarrow$ L shrinks $\rightarrow$ p_{halo} saturates $\rightarrow$ intersection mechanism fails—is consistent with the saturation mechanism predicted by the relaxed theoretical derivation above.

	V	L	`cand_cnt` (med)
P=10^5, B=1000	2	16384	1799
W=2, VL=32768	4	8192	7377
$p_{\text{halo}}(V{=}2) \approx 0.30$;	$p_{\text{halo}}(V{=}4) = 1.00$ (saturated)		

Table S17: Fixed-budget reversal example: increasing V from 2 to 4 halves L, saturating p_{halo} to 1.0 and inflating candidate cost by roughly 4×.

The asymmetry between the two dimensions is now evident: L/B directly controls the feasibility threshold of the intersection mechanism through c and exhibits a sharp transition around the **capacity threshold**—once below the threshold, candidate cost degrades sharply and is difficult to compensate; P/L, above the threshold, mainly amplifies the absolute candidate count by mildly modifying the effective c, and this amplification can be effectively absorbed by the exponential suppression of V.

S5.3 Example Verification and Implementation Details for the Runtime Stopping Rule

This subsection supplements the knee-detection implementation described in §5.4, provides post-hoc verification, and derives the resulting runtime fallback rules.

Implementation of knee detection The system sweeps upward through a preset `top_r` schedule (e.g., $\{10^{-3}, 2\times10^{-3}, 5\times10^{-3}, 10^{-2}, \ldots\}$), recording at each step the energy coverage E_{cdf} (cumulative proportion of section energy ranked in descending order) and the candidate cost `cand_cnt`. The joint knee is detected by monitoring the marginal ratio $\Delta E_{\text{cdf}}/\Delta\log(\texttt{cand_cnt})$: when this ratio drops significantly below the level of preceding steps, the system declares entry into the diminishing-returns region and terminates expansion.

Post-hoc verification Figure S19 provides offline verification by plotting candidate cost against ground-truth recall (fraction of total diff captured by the candidate set) for each `top_r` value. At small `top_r`, increasing the truncation budget improves recall at moderate candidate cost; beyond the knee, candidate cost accelerates sharply while recall gains diminish. The knee-based truncation point (r_{top}=0.25, dashed reference line) sits at the onset of this diminishing-returns region. This verification requires ground truth and is used only for post-hoc justification; online deployment relies solely on the energy-based signal.

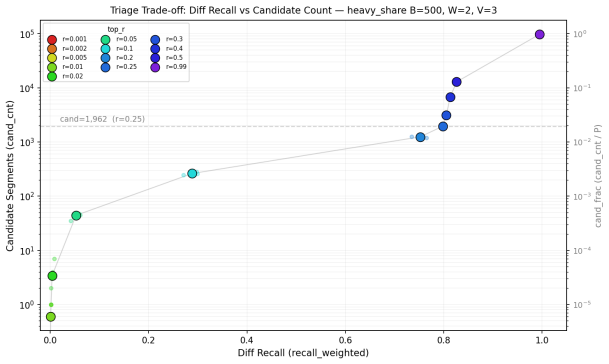


Figure S19: Recall–cost trade-off for the same configuration as Figure 6. Each colored point corresponds to a `top_r` value; the horizontal dashed line marks the knee (r_{top}=0.25). Beyond the knee, candidate cost grows rapidly while recall gains diminish.

S6 Baseline Comparison: Supplementary Details

This section provides supplementary evidence for the baseline comparison in §5.5. We first summarize budget alignment, engineering implementation details, and a capability overview in §S6.1, then supplement robustness evidence for triage and gating comparisons in §S6.2 and §S6.3, respectively; complete threshold sweep tables are provided at the end for reference. Figure S20 illustrates the hierarchical bucket tree structure of B3 for reference in subsequent discussions. Table S18 summarizes the capability comparison of all methods across the four dimensions of triage, gating, sizing, and control-plane cost.

S6.1 Budget Alignment and Capability Overview

Budget conversion Table S19 lists the digest budget conversion for each method in the triage comparison; Table S20 provides the digest budget alignment at the ∼768 KiB tier for the gating comparison.

Dimension	SCOUT	B2 (CountSketch ranking)	B3 (bucket tree)	B1 (checksum)
Triage feasible region	Wide (still usable at high `S/B` + low `ratio`)	Narrow (extensive N/A at low `ratio`)	Narrow (step-wise inflation from leaf bucket expansion)	Degrades linearly with `S/B` (no ranking)
Gating discriminability	✓ (viable operating point exists)	× (no viable trade-off)	× (no viable trade-off)	N/A (no shape signal)
Sizing	✓ (validated in §5.3)	× (per-segment ranking only)	Limited (bucket-level energy, not segment-level)	× (count only)
Control-plane cost	$O(1)$ (fixed ∼512 KiB)	$O(1)$ (same order)	$O(1)$ (same order)	$O(P)$ (∼tens of MB at $P = 10^6$)
Can scaling mitigate failure?	Partially (V is the primary knob)	Limited (core failure region remains unreachable)	Limited (bounded by structural inflation)	N/A

Table S18: Capability matrix (same budget, single-round probe): comparison across triage, gating, sizing, and control-plane cost.

Method	Key parameters	Memory formula	Estimated size
SCOUT	`V=2, L=2048, m0=32, k=3, W=2`	$V \cdot L \cdot m_0 \cdot 4$ bytes	∼512 KiB
B2-w11k	`views=2, rows=3, width=11264`	views · rows · width · 8 bytes	∼528 KiB
B2-w22k	`views=2, rows=3, width=22528`	same as above (2× width)	∼1.03 MiB
B3-c4	`b=8, depth=5, cells_per_bucket=4`	total_nodes · m · 4 bytes	∼585 KiB
B3-c8	`b=8, depth=5, cells_per_bucket=8`	same as above (2× cells)	∼1.14 MiB

Table S19: Digest budget alignment for triage comparison (primary configuration ∼512 KiB; B2-w22k and B3-c8 serve as 2× memory controls).

Method	Key parameters	Estimated size
SCOUT	`V=3, L=2048, m0=32, k=2, W=3`	∼768 KiB
B2	`views=3, rows=3, width=11264, K=2000`	∼792 KiB
B3	`b=8,depth=5,k=3,t=8,cells_per_bucket=6`	∼878 KiB

Table S20: Digest budget alignment at the ∼768 KiB tier (gating comparison).

Accumulator width and budget fairness B2 uses int64 accumulators following common industrial implementations (libraries such as Apache DataSketches and Twitter Algebird default to 64-bit accumulators). This choice is motivated by two engineering considerations. First, during the local build phase (before differencing), the single-side digest must accumulate raw counts over a full scan; when data volume reaches 10^9 and significant skew is present, a wider accumulator provides the necessary safety margin. Second, the decoding phase involves squaring accumulators and computing concentration statistics, where int64 avoids overflow risk in intermediate steps. In contrast, SCOUT's two-level mapping restricts each key's updates to the $W \cdot m_0$ cells within its window; the worst-case per-cell accumulation is bounded by the window capacity, which falls well below the int32 limit under typical parameters. Under skewed workloads (e.g., YCSB Zipfian), hot segments concentrate large numbers of keys into the same window of $W \cdot m_0$ cells. Taking the worst case of 10^9 records, k=3, $W \cdot m_0 = 4 \times 32 = 128$, the per-cell load is approximately $10^9 \times 3/128 \approx 2.3 \times 10^7$, well below the int32 limit ($\sim 2.1 \times 10^9$), providing ample safety margin.

It should be noted that B2's use of int64 yields roughly half the number of counters compared to an int32 scheme at the same budget. To rule out this discrepancy as a confound, B2's 2× memory control (B2-w22k) matches the counter count of the int32 baseline configuration.

S6.2 Triage Robustness

S6.2.1 Scaling Control and Failure Region

Figure S21 and Figure S22 confirm this conclusion from two perspectives—the two-dimensional phase diagram and per-`S/B` slices, respectively: increasing the width produces visible budget reductions at lower `S/B` or higher `ratio`, but at very high `S/B` and very low `ratio`, both width tiers approach or exceed the 2% candidate budget cap. B2's core failure region therefore persists after scaling up, consistent with the analysis in §5.5. Here B2-w11k denotes B2 with width 11,264, and B3-c4 denotes B3 with 4 cells per bucket. Numerical comparisons at representative stress points are provided in Supplement Table S21.

S/B	ratio	SCOUT	B2-w11k	B2-w22k	Notes
10	200	$\sim 2\times10^{-4}$	$\sim 1\times10^{-4}$	—	High-separability regime; B2 achieves lower candidate budget
10	2–3	$\sim 2\times10^{-4}$–7×10^{-4}	$\sim 2\times10^{-3}$	$\sim 1\times10^{-3}$	Low-separability regime; SCOUT achieves significantly lower candidate budget
100	≥ 5	$\sim 2\times10^{-4}$	$\sim 3\times10^{-3}$–1×10^{-2}	—	SCOUT quickly returns to the low-cost region
100	2–3	$\sim 7\times10^{-4}$–degraded	~ 0.02 (hitting cap)	~ 0.01–0.02	Near the indistinguishability boundary; both degrade
≥ 1000	≤ 30	partially reachable	**N/A**	few boundary points reachable	B2 exhibits a large failure surface
≥ 1000	≥ 50	$\sim 10^{-3}$	**N/A**	**N/A**	SCOUT still viable; B2 at 2× memory remains unreachable

Table S21: Threshold budget (cand_frac) comparison at representative stress points. Rows correspond to pressure regions in Figure S21; B2-w22k denotes B2's 2× memory control; "—" indicates same order as the adjacent column.

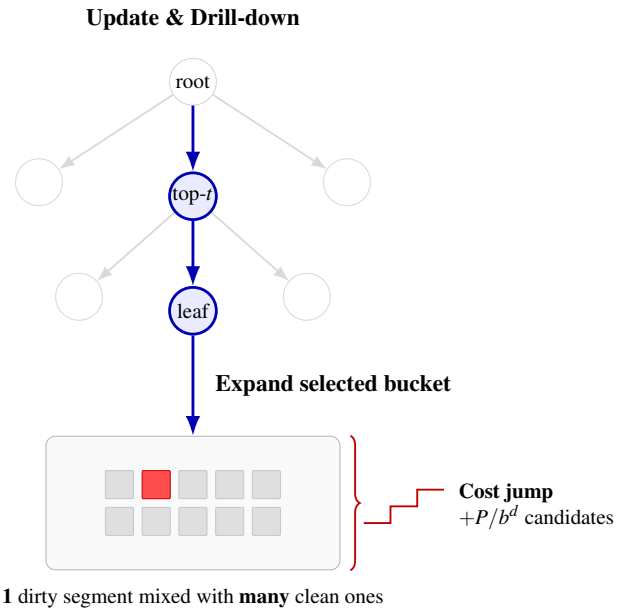


Figure S20: B3 (hierarchical bucket tree) baseline: segments are mapped to leaf nodes of a fixed tree (b=3, d=2 in this example). During reconciliation, the probe drills down from root to leaves by selecting high-score buckets at each level. Once a leaf bucket is selected, *all* segments in that bucket—including the many clean ones—must be expanded as candidates, causing a step-wise cost jump.

Slice view: per-S/B threshold budget curves To visualize the boundary shape of the failure surface more intuitively, we present the two-dimensional phase diagram from Figure 7 as slice curves. Figure S23 fixes S/B=1000 and shows each method's threshold budget as a function of ratio: B3-c4 falls below the 2% cap only in the narrow interval ratio ≥ 20; the 2× memory control (B3-c8) extends the feasible region to ratio $\approx$ 8–15, but lower ratio values remain N/A. Figure S24 further unfolds S/B into multiple subplots, focusing on the SCOUT vs. B2 comparison: as S/B increases, B2's curve rapidly approaches the budget cap, while SCOUT maintains low cost over a wider range of ratio.

S6.2.2 Relative Threshold Budget Ratio

To avoid misinterpretation from examining absolute threshold budgets alone, we additionally present the relative budget ratio (method / SCOUT). Figures S25a and S25b show the point-wise ratio for B2/SCOUT and B3-c4/SCOUT, respectively, where a higher ratio indicates that a larger candidate budget is needed to reach the same threshold; FAIL indicates the method has no feasible solution within $\leq 2\%$ while SCOUT does, ONLY indicates only that method is feasible while SCOUT is not, and N/A indicates neither is feasible.

S6.2.3 B Ablation: $B = 10 \to 100$

The main results in §5.5 are based on $B = 10$ (i.e., 10 heavily dirty segments). To examine whether the triage advantage persists when the number of heavy faults B increases, we raise B from 10 to 100 and proportionally scale total_diff (200K $\to$ 2M) to keep the per-big-segment signal strength unchanged, so that performance changes are primarily driven by capacity pressure (L/B decreasing from 204.8 to 20.48). This change approaches the capacity threshold given by Eq. (S6) in Supplement §S5.2.1 ($L/B \gtrsim 16.3$ for W=2, α=1); the

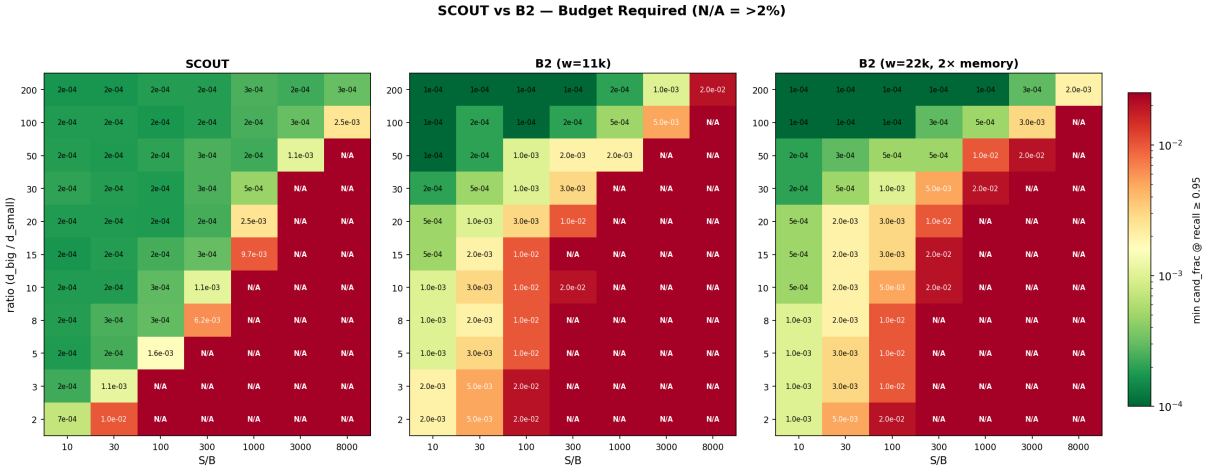


Figure S21: Three-way comparison heatmap after doubling B2's memory to 2×. Color scale matches Figure 7; N/A indicates no feasible solution within the 2% candidate budget cap.

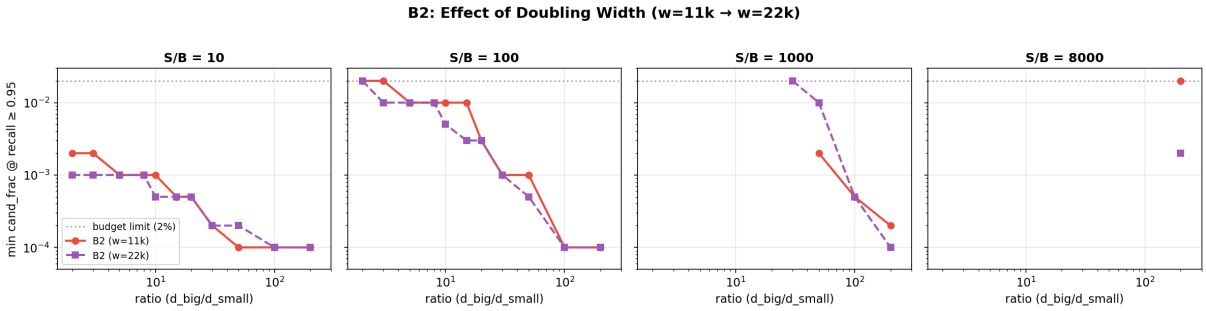


Figure S22: B2 scaling 2× slice comparison. Each subplot fixes one S/B; dashed line marks the 2% cap. In high-S/B slices, both width tiers approach or exceed the budget cap.

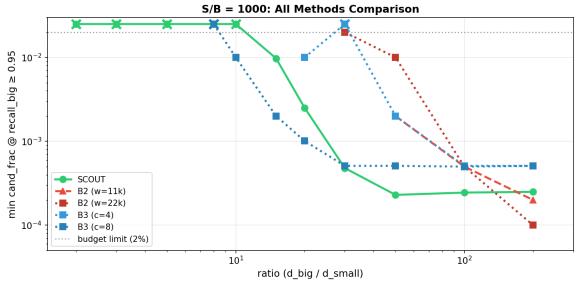


Figure S23: Threshold budget curves for all methods at the `S/B=1000` slice. Lower is better; dashed line marks the 2% cap.

widespread degradation at B=100 is therefore consistent with theoretical expectations.

The feasible-region changes on the shared grid (`S/B` $\in \{10,30,100,300\}$ $\times$ `ratio` $\in \{2,5,10,30,50,100,200\}$ = 28 points) under the 2% budget cap are shown in Table S22:

Method	$B = 10$	$B = 100$	Degradation
SCOUT	89% (25/28)	64% (18/28)	-25pp
B2(`w=11k`)	93% (26/28)	50% (14/28)	-43pp
B3(`c=4`)	93% (26/28)	54% (15/28)	-39pp

Table S22: B ablation: feasible-region coverage change (2% cap; 28-point grid).

All three methods degrade due to capacity pressure, but the degradation magnitudes differ (Table S22). Relaxing the candidate budget cap can partially recover the feasible regions of B2/B3; see the sensitivity analysis below.

S6.2.4 Sensitivity to the Candidate Budget Cap

The main text uses 2% as the candidate budget cap. A natural concern is whether this cap prematurely truncates the feasible regions of B2/B3. To address this, we relax the cap from 2% to 5% and 10% under the B=100 scenario, and on the 35-point full grid count whether each method has an operating point satisfying the recall constraint within the given cap (Table S23).

cap (`cand_frac`)	SCOUT	B2 (w=11k)	B3 (c=4)
0.02	57.1% (20/35)	40.0% (14/35)	45.7% (16/35)
0.05	57.1% (20/35)	51.4% (18/35)	57.1% (20/35)
0.10	62.9% (22/35)	60.0% (21/35)	71.4% (25/35)

Table S23: Feasible-region coverage under candidate-budget caps (B=100, 35-point full grid).

Relaxing the cap does recover some of B2/B3's N/A points: for example, at the 5% cap B3 matches SCOUT (both at 57.1%), and at the 10% cap B3 further improves to 71.4%. However, matching coverage does not equate to matching end-to-end cost. Examining the color scales in Figure S26 and Figure S27, the newly recovered points for B2/B3 have `cand_frac` values that are generally close to the cap ceiling (2×10^{-2}–10^{-1}), whereas SCOUT's typical operating points concentrate in the lower candidate range (5×10^{-3}–10^{-2}). The 2% cap therefore acts more as a practical downstream cap: it barely constrains SCOUT's natural operating range, but significantly affects whether B2/B3 remain feasible only near the cap ceiling at high-pressure stress points.

Furthermore, B3's recovery under relaxed caps is more pronounced, consistent with the discrete granularity of its leaf-bucket expansion: under a strict cap, step-wise jumps are more likely to hit the ceiling directly, while a relaxed cap more readily accommodates these jumps.

S6.3 Gating Robustness

S6.3.1 Proxy Gating Score Definitions

For each method, we select a "diffuseness" proxy score consistent with its digest structure, and uniformly sweep the threshold τ_{gating} to report FAR–FRR and the decisive classification fraction. This section supplements the specific definitions and settings of each score.

SCOUT uses K_{eff}/L from the section energy spectrum. B2 uses the head-mass residual $1-\rho_{\text{topK}}$, where

$$\rho_{\text{topK}} = \frac{\sum_{p\in\text{topK}} \hat{d}_p}{\sum_p \hat{d}_p}.$$

The intuition is that concentrated faults are dominated by a few segments, so $\rho_{\text{topK}} \approx 1$ and $1-\rho_{\text{topK}}$ is close to 0; under diffuse discrepancies, the top-K explains only a small fraction of total mass, and $1-\rho_{\text{topK}}$ rises. For consistency with the candidate budget criterion, we bind K to the 2% candidate cap. B3 uses the hierarchical expansion ratio

$$\text{expansion_score} = \max_{\ell\in\{1,\ldots,\text{depth}\}} \frac{\text{selected_buckets}_\ell}{\text{total_buckets}_\ell},$$

with a fixed per-level selection strategy `top_t_per_level=8`; the motivation is to align the score with B3's coarse-to-fine navigation process: the more diffuse the discrepancy, the more branches must be expanded to maintain recall.

We also considered alternative candidate scores. For B2, the Gini coefficient and top-K/top-$2K$ mass ratio are both based on the sorted estimated distribution and are highly collinear with $1-\rho_{\text{topK}}$, offering no qualitative improvement in discriminability. For B3, the average expansion ratio is diluted by the large number of unexpanded levels, losing the signal, while the discrete count of expanded levels produces more step-like FAR–FRR curves. The common limitation of these alternative scores is that B2 through hash aliasing and B3

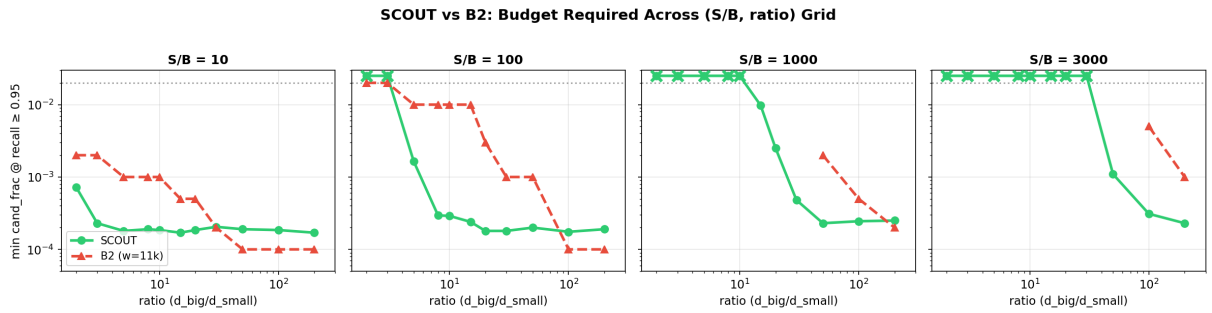
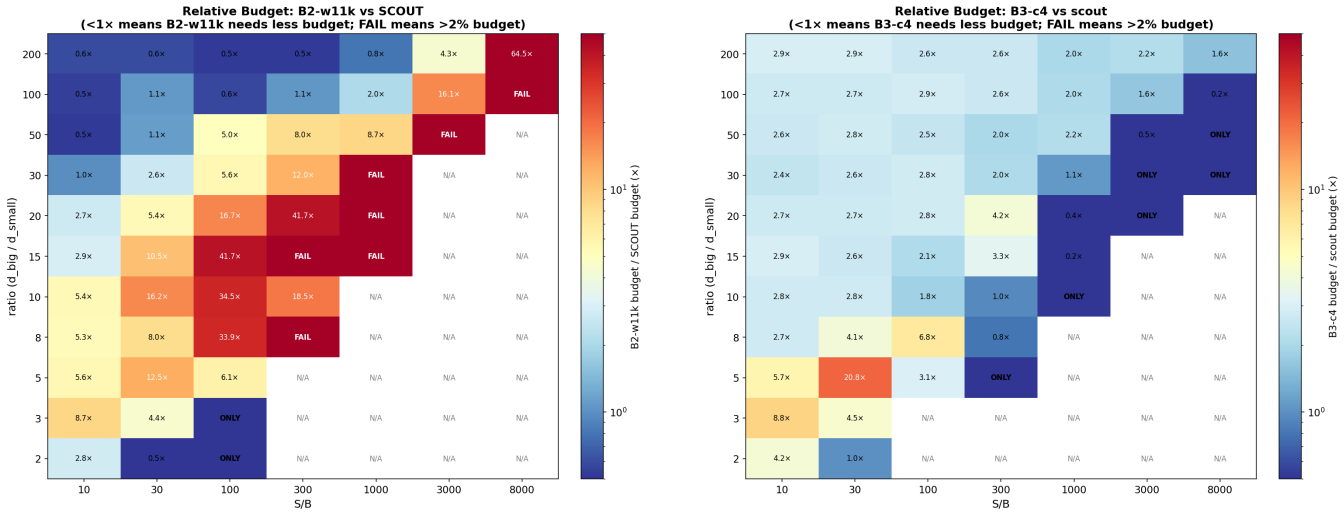


Figure S24: SCOUT vs. B2 per-`S/B` slice comparison. Each subplot fixes one `S/B`; Lower is better; dashed line marks the 2% cap.



(a) B2 / SCOUT relative threshold budget ratio. `FAIL`: B2 has no feasible solution within the $\leq 2\%$ cap while SCOUT does; `ONLY`: only B2 is feasible; `N/A`: neither is feasible.

(b) B3-c4 / SCOUT relative threshold budget ratio; `FAIL`/`ONLY`/`N/A` have the same meaning as in the left panel (with the comparison target replaced by B3-c4).

Figure S25: Relative threshold budget ratio heatmaps. Each cell shows the minimum `cand_frac` needed to achieve `median(recall_big)` ≥ 0.95, relative to SCOUT (> 1 indicates higher cost).

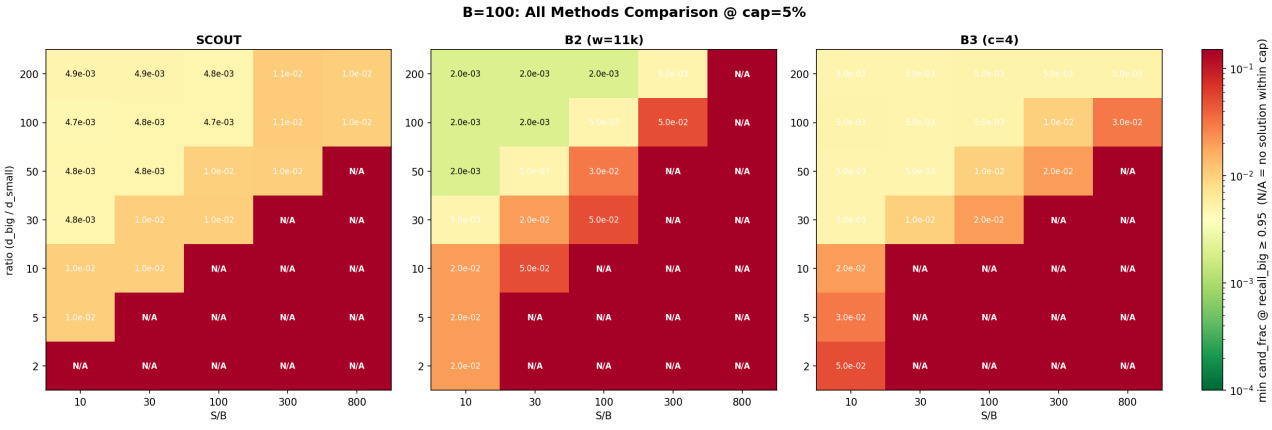


Figure S26: Three-method heatmap at cap=5% (B=100). Color scale matches Figure 7; N/A indicates no feasible solution within the 5% candidate budget.

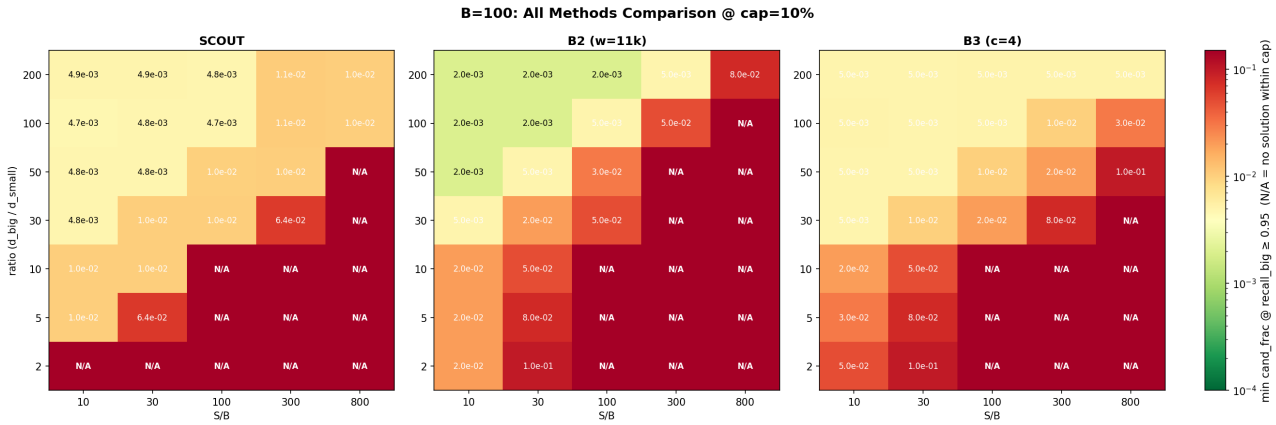


Figure S27: Three-method heatmap at cap=10% (B=100). Color scale matches Figure 7; N/A indicates no feasible solution within the 10% candidate budget.

through hierarchical aggregation both destroy spatial locality at the digest construction stage; any score based on such representations amounts to post-hoc inference on a representation that has already lost shape information, and the ceiling on discriminability is limited by the representation itself, not by the choice of score.

S6.3.2 Budget Control: ∼768 KiB Tier

The main results in §5.5 are based on the ∼512 KiB tier, matching the triage budget. To verify that conclusions do not change with budget, this section adds a control at the ∼768 KiB tier (SCOUT: `V=3/W=3`; B2: `views=3`; B3: `cells=6`); results are shown in Figure S28 and Table S24. Conclusions are consistent across both budget tiers: SCOUT's optimal FAR is virtually unchanged (∼0.038 vs. ∼0.039), while B2's `FRR` = 0.50 plateau and B3's `FAR` ≈ 1.0 phenomenon both persist regardless of budget, indicating that the failures originate from structural limitations.

S6.3.3 Gating-Curve Shape and FAR Source

The FAR–FRR curves of B2 and B3 in Figure S28 exhibit plateaus or steps. The cause is that the mix workload samples the transition zone with a small number of discrete noise-coverage tiers; the scores from each sub-scenario cluster into narrow bands, and when the threshold crosses a cluster boundary it triggers a batch flip. This phenomenon does not affect the main conclusions.

Specifically, at the operating point τ_{gating}=0.40, FAR is approximately 4%. Stratifying by noise coverage n (Figure S29), the FAR is almost entirely attributable to the n=0.3 subset (approximately 30% of segments containing background noise), while the FAR of all other subsets is near zero: the global shape of this boundary scenario already closely resembles the diffuse type, and SCOUT classifying it as diffuse constitutes a borderline misclassification.

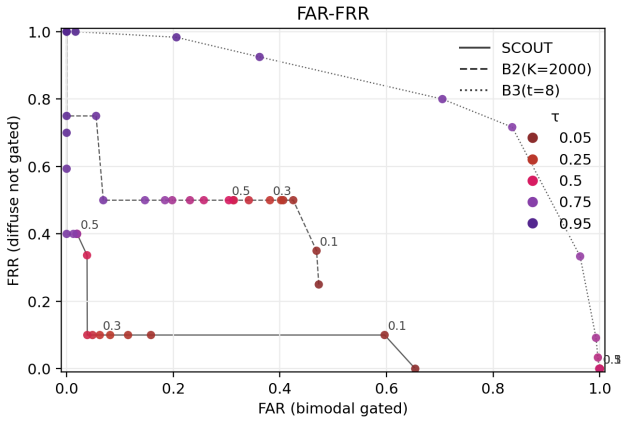


Figure S28: Gating FAR–FRR trade-off curve (∼768 KiB, same budget). Each point on a curve corresponds to a gating threshold τ_{gating} (annotated alongside the curve); smaller τ_{gating} is more aggressive (FAR rises, FRR falls).

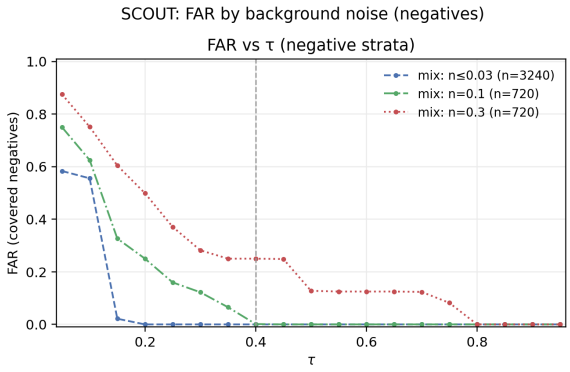


Figure S29: FAR stratified by noise coverage n. Only the $n = 0.3$ subset (approximately 30% of segments containing background noise) contributes nearly all of the FAR; the FAR of all other subsets is close to zero.

Method	τ_{gating}	FAR_cond	FRR_cond	decisive (neg)
SCOUT	0.30	0.062	0.10	0.93
SCOUT	0.35	0.049	0.10	0.93
SCOUT	**0.40**	**0.039**	**0.10**	0.93
SCOUT	0.45	0.038	0.34	0.93
B2	0.05	**0.47**	0.25	1.0
B2	0.15	0.43	**0.50**	1.0
B2	0.35	0.34	**0.50**	1.0
B2	0.80	0.07	**0.50**	1.0
B2	0.90	0.00	0.75	1.0
B3	0.50	**1.00**	0.00	0.875
B3	0.65	**0.99**	0.09	0.875
B3	0.70	**0.96**	0.33	0.875
B3	0.85	0.36	0.93	0.875
B3	0.95	0.02	1.00	0.875

Table S24: Complete threshold sweep for the ∼768 KiB control setting (`min_touched=64`). The table reports conditional false-alarm rate `FAR_cond`, conditional false-reject rate `FRR_cond`, and `decisive (neg)`. Here, `decisive (neg)` denotes the fraction of negative-class samples that receive a decisive (non-`unknown`) classification; the negative class corresponds to bimodal/mix samples, since `gate` is treated as the positive class.

S6.3.4 `unknown` Fraction Does Not Affect Conclusions

For gating, the main text reports FAR_{cond} and FRR_{cond} over decisive samples, excluding samples labeled `unknown`. Supplement Table S24 therefore reports `decisive (neg)`, the fraction of negative-class samples that receive a decisive rather than `unknown` classification. The reported values are 0.93 for SCOUT, 1.0 for B2, and 0.875 for B3, which correspond to approximately 7%, 0%, and 12.5% `unknown`, respectively. These exclusion rates are modest and do not account for the observed errors: B2 still reaches $\text{FAR}_{\text{cond}} = 0.47$ with 100% decisive classifications, and B3 remains near $\text{FAR}_{\text{cond}} = 1.0$ at $\tau_{\text{gating}} = 0.50$. The failures therefore reflect insufficient discriminability rather than the exclusion of difficult samples as `unknown`.

S7 YCSB Validation: Supplementary Details

This section supplements the experimental setup and extended results from §5.6.

Data scale and segment size skew Table S26 summarizes the segment size distribution under YCSB Zipfian (α=0.99) at three data-scale tiers. As the number of records grows from 20M to 2×10^9, the gap between the hottest and coldest segments widens to nearly 50×, reflecting the typical structure of Zipfian skew under range partitioning.

Fault families and parameterization The injection methods and severity sweep parameters for the four fault families are given in Table S25. Bimodal and Drift-shaped target concentrated patterns: the former drops a fixed fraction of keys within a contiguous window, while the latter concentrates additive discrepancies on the hottest segments, modelling replica lag under Zipfian skew. Uniform Diffuse and Weighted Diffuse both target diffuse patterns; Uniform Diffuse samples non-empty segments uniformly, whereas Weighted Diffuse samples dropped keys with probability proportional to each segment's record count, simulating independent per-record loss.

Sizing accuracy Drift-shaped is the most demanding fault family for sizing robustness, as it combines segment-size skew with a workload-shaped concentration of discrepancies in hot segments. With the Bessel correction applied, the median |rel_err_scaled| of the de-drifted estimator converges to approximately 1% across all three data-scale tiers, matching the accuracy observed under the uniform settings of §5.3. Per-tier values are reported in Table S27.

Weighted Diffuse and the head-triage interpretation Weighted Diffuse is deferred from the main text because the standard `recall_big` metric does not apply directly: unlike Bimodal and Drift-shaped, whose injection geometry provides a natural heavy-segment set, Weighted Diffuse spreads discrepancies across nearly all segments while concentrating most diff mass in a small head (the top 285 segments, 0.3% of P, account for 50% of total diff mass). We therefore define a mass-based head target $H_{0.5}$—the minimum prefix covering 50% of diff mass when segments are sorted by discrepancy count—and report two companion metrics: *head recall* ($|C \cap H_{0.5}|/|H_{0.5}|$), the fraction of heavy segments captured by the candidate set, and *head mass coverage* ($\sum_{p\in C\cap H_{0.5}} d_p / \sum_{p\in H_{0.5}} d_p$), the share of their discrepancy mass that is captured. The latter is typically higher because top-ranked segments each contribute disproportionately more mass.

Table S28 summarizes the resulting operating points. At a candidate fraction of approximately 0.009 (i.e., scanning less than 1% of all segments), SCOUT already covers roughly 83% of the head mass; at approximately 0.028, coverage rises to about 90%. These operating points are nearly invariant across the three data-scale tiers, consistent with the stability observed for the other families in the main text.

S8 Hash Quality and Sensitivity to k

This paper treats each view as an independent hash channel. In our implementation and evaluation, this independence is enforced through separate randomization across views in both the mapping from segments to window start positions and the

Fault family	Parameterization
Bimodal	Proportional key dropping within a contiguous 3-segment window, $r_{\text{drop}} \in \{0.05, 0.1, 0.2, 0.5\}$.
Drift- shaped	Additive discrepancies concentrated on the hottest segments (r_{add}=0.005), modelling replica-lag under skew.
Weighted Diffuse	Global random key dropping, with dropped keys sampled with probability proportional to each segment's record count; $r_{\text{drop}} \in \{0.001, 0.003, 0.01, 0.03\}$. This simulates independent per-record loss under segment-size skew.
Uniform Diffuse	Global random key dropping over non-empty segments with equal probability; $r_{\text{drop}} \in \{0.001, 0.003, 0.01, 0.03\}$.

Table S25: Fault families and their parameterization (main text §5.6).

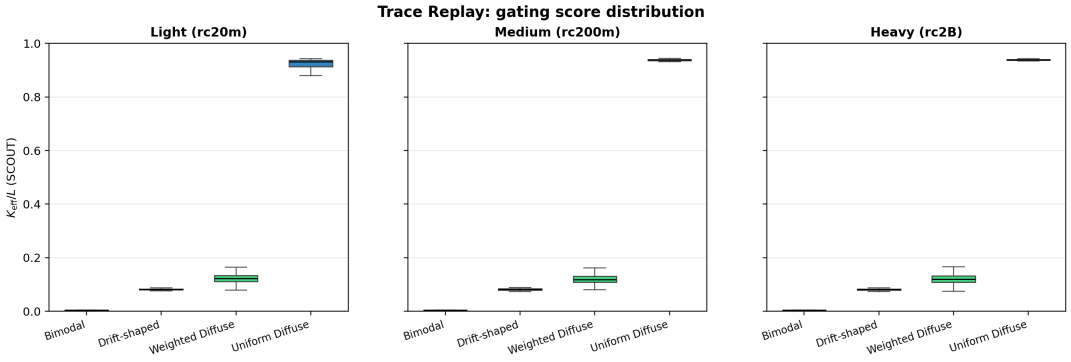


Figure S30: Distribution of K_{eff}/L under YCSB-driven workloads. Each panel corresponds to one data-scale tier; the four box-plot groups correspond to the four fault families.

Density tier	Records	seg size p50	p99
Light	20M	35	1,689
Medium	200M	349	16,730
Heavy	2B	3,487	167,675

Table S26: The three density tiers correspond to the segment fill states of the same YCSB Zipfian workload at different data volumes.

Density	dedrift med	dedrift p90	raw med	raw p90
Light	0.0077	0.0175	0.0397	0.0614
Medium	0.0141	0.0280	0.0313	0.0431
Heavy	0.0096	0.0225	0.0384	0.0516

Table S27: Sizing error (|rel_err_scaled|) under Drift-shaped faults. The de-drift estimator outperforms the raw baseline across all three data-scale tiers.

	head recall			head mass cov.		
`cand_frac`	Light	Medium	Heavy	Light	Medium	Heavy
$\sim 10^{-5}$	0.004	0.004	0.004	0.15	0.16	0.16
$\sim$0.009	0.40	0.39	0.40	0.83	0.83	0.84
$\sim$0.028	0.61	0.60	0.60	0.90	0.90	0.90
$\sim$0.127	0.88	0.89	0.88	0.97	0.98	0.98

Table S28: Weighted Diffuse head-triage operating points (SCOUT, $H_{0.5}$ target). Values are representative averaged across four severity levels (r_{drop}).

k within-window hash edges used by each key. To preserve the "k distinct cells" semantics assumed in the main text, the system retries until it obtains k distinct landing positions for each key, and falls back to deterministic placement only if the retry budget is exhausted.

S8.1 Key Input Structure and Hash Quality

The preceding evaluation defaults to a high-grade hash (`sha256`). This section examines whether the triage operating point changes when hash quality is degraded or key inputs deviate from a uniform distribution.

The experiment sweeps two key structures—`sequential` (consecutively incrementing keys) and `hotset` (keys with a

k	raw p90 (|rel_err_scaled|)	corrected p90 ($\|\cdot\|$)
1	0.0184	0.0184
2	0.0197	0.0189
3	0.0207	0.0185

Table S29: Scaling bias and post-processing correction for varying k.

localized hotspot)—and three hash implementations: `sha256`, `mix64_only` (integer mixing only), and `raw_mod` (direct modular reduction, used here as a weak-hash negative control). The operating point is taken as the lowest-budget point achieving `succ` ≥ 0.95, with the median (p50) and 90th-percentile (p90) candidate fraction reported at that point.

Table S30 reports representative configurations. Across all of them, the effect of hash implementation and key structure remains within measurement noise: the discrete-grid best `top_r` values are identical, and `cand_p50`/`cand_p90` differ by no more than 5%. Even the negative control `raw_mod` does not produce an observable shift within the tested parameter range. SCOUT's triage operating point is therefore largely insensitive to hash implementation and key distribution pattern.

S8.2 Effect of k on Triage and Sizing

We next examine how varying k affects triage and sizing. For triage, the fixed-budget envelope from the $m_0 \times k$ sweep shows no reproducible separation among the budget–reliability curves for $k \in \{1,2,3\}$ over the tested range.

Table S31 shows the corresponding step-like behavior under a fixed candidate budget: a slice is either clearly below the usable budget range or already in the stable regime, and the transition occurs at essentially the same budget for all three k values.

The effect of k is more visible in sizing. Increasing k changes the estimator through two mechanisms. The first is locality bias (§S4.2): larger k increases k/M_{local}, thereby amplifying the discrepancy between the global approximation $(1-k/M)$ and the local reality $(1-k/M_{\text{local}})$. The second is diminishing marginal information gain at the section level: because the sizing statistic E_s is computed per section over m_0 cells, the information contributed by additional updates becomes mildly sublinear as k grows. To capture this second effect, we use

$$k_{\text{eff}}(m_0,k) = m_0\left(1-(1-\tfrac{1}{m_0})^k\right)$$

to approximate the effective number of occupied cells in a section-level occupancy model with k throws. This yields a corresponding correction factor, k/k_{eff}, for the scaling bias. Table S29 summarizes the p90 sizing error across three levels of k, aggregated over all configurations and `top_r` values. The raw p90 (with Bessel correction only) increases monotonically with k. After applying the k/k_{eff} correction, the corrected p90 falls into a narrow band of 1.84%–1.89%, with mutual differences below 0.1 percentage points. Once this systematic trend is removed, the residual is dominated by sampling variance and no longer shows a monotonic dependence on k.

Taken together, these results show that, within the operating range covered by this paper, hash implementation and key structure do not produce an observable shift in the triage operating point. The main effect of varying k appears instead in sizing, where it introduces a predictable scaling bias that is substantially aligned by the k_{eff} correction factor.

S9 Notations

Core probe geometry parameters are listed in Table 2; the system model, algorithm definitions, and evaluation metrics are given in §2, §4, and §5.1, respectively. This section supplements the remaining symbols.

Config (P,L,B)	hash_mode	key distribution pattern	best top_r	cand_p50	cand_p90
$(10^5,2048,10)$	sha256	sequential	0.02	1.10e-4	1.20e-4
$(10^5,2048,10)$	sha256	hotset	0.02	1.10e-4	1.20e-4
$(10^5,2048,10)$	mix64_only	sequential	0.02	1.10e-4	1.20e-4
$(10^5,2048,10)$	mix64_only	hotset	0.02	1.10e-4	1.20e-4
$(10^5,2048,10)$	raw_mod	sequential	0.02	1.10e-4	1.20e-4
$(10^5,2048,10)$	raw_mod	hotset	0.02	1.00e-4	1.20e-4
$(10^7,8192,10)$	sha256	sequential	0.005	1.10e-6	1.30e-6
$(10^7,8192,10)$	mix64_only	sequential	0.005	1.10e-6	1.30e-6
$(10^7,8192,10)$	raw_mod	sequential	0.005	1.10e-6	1.30e-6
$(10^5,4096,100)$	sha256	sequential	0.10	1.98e-3	2.10e-3
$(10^5,4096,100)$	mix64_only	sequential	0.10	1.97e-3	2.11e-3
$(10^5,4096,100)$	raw_mod	sequential	0.10	1.95e-3	2.05e-3

Table S30: Triage comparison under different key structures and hash qualities (representative configurations).

Slice family	Envelope summary at representative budgets
P=10^6, L=8192, B=100, d_{big}=200, $m_0 \in \{64,128,256\}$	At cand_frac $\le 2\times10^{-4}$, all $k \in \{1,2,3\}$ fail (succ@0.95 $= 0$); at $\le 5\times10^{-4}$ and $\le 10^{-3}$, all succeed (succ@0.95 $= 1$).
P=10^6, L=32768, B=100, d_{big}=200, $m_0 \in \{64,128,256\}$	All three k values already achieve succ@0.95 $= 1$ at cand_frac $\le 2\times10^{-4}$; the envelope remains identical at larger budgets.
P=10^3, L=65536, B=10, d_{big}=200, $m_0 \in \{64,128,256\}$	All three k values remain below the reliability target even at cand_frac $\le 10^{-3}$, giving succ@0.95 $= 0$ throughout.

Table S31: Representative fixed-budget triage envelopes from the $m_0 \times k$ sweep across three slice families.

Symbol	Meaning
Differences and workloads	
D^+, D^-	Number of diff keys present only on side A (B)
$\hat{D}$	Scale estimate output by sizing
S	Number of noise segments in mixed workloads (each carrying d_{small} differences)
d_{big}	Difference intensity of a single heavily dirty segment
d_{small}	Difference intensity of a single noise segment
ratio	Separability knob, ratio $= d_{\text{big}}/d_{\text{small}}$
S/B	Noise-to-signal segment multiplier (count ratio in mixed workloads)
q	Drift parameter: probability that a diff key belongs to side A (side_plus_prob)
Probe geometry	
M	Total cell count of a single view, $M = L \cdot m_0$
M_{bytes}	Probe digest size in bytes, $M_{\text{bytes}} = 4VLm_0$
$h_v(p)$	Window-start hash of segment p in view v

Table S32: Symbol table (I): differences and probe geometry.

Symbol	Meaning
Triage and operating points	
Ω_u	Window aggregation score, $\Omega_u = \sum_{s=u}^{u+W-1} E_s$
t	Truncation window count, $t = \lceil r \cdot (L-W+1) \rceil$
r_{top} (top_r)	Truncation ratio knob for triage
p_{halo}	False-inclusion probability of a clean segment in a single view
$\mathcal{C}$	Candidate segment set
cand_cnt	Candidate segment count (i.e., $\lvert\mathcal{C}\rvert$)
cand_frac	Candidate fraction, cand_frac $= \lvert\mathcal{C}\rvert/P$
recall_big	Fraction of d_{big} segments covered by the candidate set
Baselines	
ρ_{topK}	B2 head-mass fraction, $\rho_{\text{topK}} = \sum_{p\in\text{topK}} \hat{d}_p \,/\, \sum_p \hat{d}_p$

Table S33: Symbol table (II): triage, operating metrics, and baselines.

Symbol	Meaning
Section-level statistics	
Δ_j	Difference count of cell j, $C_j^{(A)} - C_j^{(B)}$
net_sum_s	First moment of section s, $\sum_{j \in s} \Delta_j$
sq_sum_s	Raw second moment of section s, $\sum_{j \in s} \Delta_j^2$
E_s	De-drifted energy, $E_s = \text{sq_sum}_s - \frac{(\text{net_sum}_s)^2}{m_0}$
$E_s^{(v)}$	De-drifted energy of section s in view v
w_s	Normalized energy weight, $w_s = E_s / \sum_s E_s$
Gating	
K_{eff}	Effective number of sections (reciprocal of the Herfindahl index), $K_{\text{eff}} = 1/\sum_s w_s^2$
K_{eff}/L	Normalized support dispersion of the energy spectrum (low = concentrated, high = diffuse)
occ	Cross-view fraction of non-empty sections (Eq. (3))
τ_{occ}	Occupancy threshold (three-state abstention boundary)
τ_{gating}	Gating concentration threshold (lower = more aggressive)
FAR_{cond}	Conditional false-alarm rate (fraction of decisive samples in which a concentrated pattern is misclassified as diffuse)
FRR_{cond}	Conditional false-rejection rate (fraction of decisive samples in which a diffuse pattern is misclassified as concentrated)

Table S34: Symbol table (III): section-level statistics and gating.